



LUX INDUSTRIES

IP Enforcement Memorandum

Prepared for: Cyrus Pahlavi, President & CEO

Hunter Dupont, Managing Partner — Strategic Transactions

Alan Donohue, General Counsel

Prepared by: Tridib Nandy — CTO, Lux Industries Inc.

2026-05-25

MEMORANDUM

TO: Cyrus Pahlavi, President & Chief Executive Officer, Lux Industries Inc.
Hunter Dupont, Managing Partner — Strategic Transactions (deal lead)
Alan Donohue, General Counsel (legal lead)

CC: Board of Directors, Lux Industries Inc.

FROM: Tridib Nandy — Chief Technology Officer, Lux Industries Inc.; concurrently
Vice President of Engineering, the Counterparty

DATE: 25 May 2026

RE: Misappropriation of Lux foundational technology by the Counterparty (Satschel, Inc. d/b/a Liquidity.io, and operating affiliate Liquidity.io LLC); legal options, damages analysis, and proposed cross-license and settlement framework

PRIVILEGE: Confidential. Attorney-client privileged. Attorney work product. Prepared in anticipation of litigation and as a board strategy document.

Author's disclosure of interest: I serve concurrently as Chief Technology Officer of Lux Industries Inc. and Vice President of Engineering at the Counterparty entity referenced herein. This memorandum is written from the Lux chair, in service of the Lux board's decision-making. My Counterparty role gives direct operational visibility into the matters described; I have set them out as observed, with the understanding that any Counterparty-side response will be developed by its own counsel and that Lux's outside litigation counsel will need to develop independent record support for any allegations herein before filing.

I. Executive Summary

The Counterparty has commercially deployed material portions of Lux Industries Inc.'s patent-protected and source-available proprietary technology since at least 2014, in production, on three live blockchain networks, and in support of a registered ATS/BD/TA business that is currently raising approximately \$75 million at a \$1.1 billion valuation, with Forum Markets (a publicly-traded company, formerly Ethzilla; ticker FRMM (NASDAQ; CIK 0001690080; principal office Palm Beach, FL; SIC 6199)) continuing as lead investor from the prior round and verbal commitments for at least approximately half the round secured to date — in each case without executing the commercial license that the governing Lux Ecosystem License v1.2 ("LEL") and the Lux Research License with Patent Reservation v1.0 ("LRL-PR") require.

Lux's direct investment in the misappropriated technology base exceeds \$20 million in development cost recorded since 2014, plus a multi-year contribution from senior engineering personnel including the Lux Chief Cryptographer and the named author of this memorandum.

The most striking evidentiary fact in this memorandum is the 90-day commit-ratio analysis at §VI: the Lux Founder and Chief Cryptographer contributed **5,991 commits** and approximately **93 million lines of code** (additions + deletions) across six venture trees — the Counterparty's primary codebase plus five related ventures (VCC, MLC, EquityTable, OnyxPlus, and the legacy/archive codebase including the simplici-* repositories) — during the most recent 90-day window, and **zero commits to Lux's upstream repositories** in the same window. The Counterparty reached its \$1.1B current-round valuation during that same 90 days. The implied

value-creation rate of the Lux Founder’s diverted bandwidth is approximately **\$12.2M per day**.

Discovery posture. This matter was identified and elevated to the Lux Industries Inc. board by Mr. Tridib Nandy (Lux CTO; concurrent VPE at the Counterparty), whose dual-role visibility — luxfi/* GitHub-org admin on the Lux side combined with operational seat on the Counterparty’s engineering organization — positioned him to observe the unauthorized integration of Lux modules into the Counterparty’s commercial stack and the absence of a corresponding executed license. Mr. Nandy remains in his VPE role; his team’s continued involvement is required for the Counterparty’s long-term operational success, and the desired outcome of this matter is an **equitable settlement preserving Counterparty operational continuity while recognising Lux’s contribution**, rather than scorched-earth litigation.

This memorandum sets out the legal record, identifies twelve causes of action available to Lux, quantifies damages at small / medium / big scenarios (\$80M to \$590M for IP claims; \$180M to \$890M with opportunity-cost component added), projects forward enterprise value at user-scale milestones up to 100M users (Lux’s equity stake value at the rebased settlement target ranges into the \$B+ at scale), analyzes the “wait vs. settle” time-value question, presents a settlement triangulation matrix the deal lead (Mr. Dupont) can use to negotiate, and locks the mandatory terms the President & CEO (Mr. Pahlavi) requires in any executed agreement.

Posture: partnership, not litigation. The objective of this memorandum is a **partnership agreement that regularises the underlying IP relationship and creates a long-term commercial alliance** between Lux Industries Inc. and the Counterparty. Litigation is the alternative path available if no partnership can be reached; the twelve available causes of action (§V) and litigation-cash damage ranges (§VI, \$80M–\$890M) are documented as **leverage and as the alternative posture**, not as the operative ask. The corporate-lawyer framing assumes both sides benefit from a deal that regularises the IP, preserves Counterparty operational continuity, and aligns Lux as a substantive partner in the Counterparty’s forward enterprise value.

Headline partnership range (equity-share). Lux’s partnership-target range is **10% to 50% equity in the Counterparty**, structured as a negotiated package of equity + cash + reciprocal commercial commitments. The 10% floor reflects the minimum substantive partnership Lux will accept; 25% is the standard landing zone; 35% broadens to a deeper commercial alliance; 50% is the full equal-partner recapitalisation. **Higher equity tiers carry escalating governance rights** (board observer → board seat → governance parity) and broader reciprocal-licence packages.

Cash and equity are inversely related to a constant total-value target. Target total nominal value = \$550M (= 50% of the \$1.1B valuation = the max-equity tier’s all-equity satisfaction of Lux’s claim). Cash fills the gap at lower equity tiers: **cash = (50% – agreed equity %) × \$1.1B**. So 50% equity ⇒ \$0 cash; 35% ⇒ \$165M cash; 25% ⇒ \$275M cash; 10% ⇒ \$440M cash. Every tier delivers the same \$550M total nominal value to Lux. The Counterparty chooses where in the range to land based on its preference between existing-shareholder dilution and cash outflow.

Substantiation at the \$1.1B valuation. The current \$75M raise being subscribed at \$1.1B valuation (Forum Markets, lead) is the natural pricing comp: equity issued to Lux at the same valuation is priced on the same per-share basis as the new-money investors, which is the clean precedent transaction a corporate lawyer would anchor on. See §X for the full tier matrix with governance + reciprocal-commitment packages and the multi-year cash sequencing that makes the lower-equity tiers practical for the Counterparty.

Mandatory term: Mr. Zach Kelling stays as Counterparty CTO (key-person retention). Mr. Kelling — formerly Lux Founder / Architect / Chief Cryptographer, now Counterparty CTO — is a prospective named defendant under the available Counts (§V) but is the lead key-person retention target in any partnership outcome. Lux’s equity stake is rendered substantially worthless without his continued technical leadership; the partnership therefore pairs mutual releases of Mr. Kelling individually with a multi-year key-person retention commitment from the Counterparty. See §XI for the full mandatory-terms list.

This is not a single-proposal document. It presents data, damage ranges, partnership-upside ranges, and negotiating floors. The deal team triangulates to a specific outcome. Mr. Pahlavi retains final approval authority on any executed agreement.

II. Parties and Dramatis Personae

Lux Industries Inc. — Decision Chain

- **Cyrus Pahlavi** — President & Chief Executive Officer. Final approval authority on any settlement, license, or litigation decision.
- **Hunter Dupont** — Largest investor in Lux Industries Inc.; Managing Partner, Strategic Transactions. Leads deal discussions and commercial negotiations with the Counterparty.
- **Alan Donohue** — General Counsel. Leads legal strategy and definitive-document drafting; provides legal support to Mr. Dupont throughout the negotiation.
- **Tridib Nandy** — Chief Technology Officer of Lux Industries Inc.; `luxfi/*` GitHub-org admin. Author of this memorandum. Holds concurrent role as Vice President of Engineering at the Counterparty (**role currently active**; his team is required for the Counterparty’s long-term operational success). It was through this concurrent visibility that Mr. Nandy identified the unauthorized integration of Lux modules into the Counterparty’s commercial stack and elevated the matter to the Lux Industries Inc. board, occasioning this memorandum.

*Mr. Zach Kelling — formerly Founder, Architect, and Chief Cryptographer of Lux Industries Inc., and formerly a `luxfi/*` GitHub-org admin — has **departed Lux Industries Inc. and is currently Chief Technology Officer of the Counterparty (Liquidity.io)**. He is therefore a **Counterparty officer for purposes of this memorandum** and is named and discussed in the Counterparty Officer Slate (below) and in §IV.E. As a former Lux Founder/officer and current Counterparty officer who took the Lux-origin IP rebuild across, Mr. Kelling is a **prospective named defendant** in the available causes of action (§V), and is the lead key-person retention target in any partnership outcome (§XI).*

Counterparty — Officer Slate

- **Zach Kelling** — **Chief Technology Officer of the Counterparty (Liquidity.io)**. Formerly Founder, Architect, and Chief Cryptographer of Lux Industries Inc.; formerly held `luxfi/*` GitHub-org admin. **Departed Lux Industries Inc. and took the CTO role at the Counterparty**, where he and a substantially overlapping team rebuilt the Counterparty’s ATS / BD / TA stack from the ground up importing the LEL- and LRL-PR-tier Lux modules (see §IV(C), (F)). His operational authority covers the platform rebuild and ongoing technical posture. **Mr.**

Kelling is NOT named in any contemplated litigation by Lux Industries Inc. The officers with the affirmative duty to identify and remediate the unlicensed-IP / OSS-licence-review failure operate *above* the Counterparty CTO role:

- Mr. Sonsurkar (CSO) bore the software-supply-chain-integrity and third-party-IP-compliance duty by job function.
- Mr. Trombley (CSO/Strategy, previously Acting CEO) bore strategic and investor-representation responsibility for the underlying technology stack.
- Mr. Choi (CEO) bore ultimate disclosure responsibility for representations to investors — though, on information and belief, he was **not informed** of the LEL / LRL-PR licensing mismatch by Mr. Sonsurkar or Mr. Trombley despite their respective duties, which mitigates his culpability from knowing fraud toward an officer-level disclosure failure absent personal knowledge.

The technical executor of the rebuild (Mr. Kelling) is not the officer the litigation theory of the case names; the licensure- review duty travels with the CSO seat and the disclosure duty travels with the CEO/strategy chain. **Mr. Kelling’s continued tenure as Counterparty CTO is a Lux-required mandatory term** in any partnership outcome (§X, §XI) — the equity Lux takes is rendered substantially worthless if the Counterparty loses Mr. Kelling’s technical leadership, because the current Counterparty operational team without him is not viewed as competent to maintain the platform. **See §IV.E.**

- **Eric Choi** — Chief Executive Officer of the Counterparty. Top operational authority; ultimate responsibility for the Counterparty’s representations to investors and regulators regarding IP ownership and licensing.
- **Austin Trombley** — Chief Strategy Officer; previously Acting Chief Executive Officer. Blockchain / AI engineer. Built a **prior** (not current) ATS infrastructure at a prior employer (Franklin Templeton) and **served as an eyewitness in the Franklin Templeton litigation** arising from that engagement. **Counsel should review the Franklin Templeton litigation record** for (i) pattern-of-conduct evidence relevant to the present matter, (ii) any limits on or adverse characterizations of his prior testimony that could be impeachable, and (iii) the technical scope and quality of the ATS he built there.

Material context: prior to Mr. Kelling’s arrival as active Chief Technology Officer at the Counterparty, the Counterparty’s ATS as built under Mr. Trombley’s technical direction **lacked even basic time-price priority order matching**, the foundational principle of any functioning alternative trading system. The pre-Kelling technical state is addressed in §IV(F).

- **Coleman Church** — Chief Executive Officer of the Counterparty’s broker-dealer affiliate. FINRA-regulated responsibility for the BD’s technology and operational integrity.
- **Jayant Sonsurkar** — Chief Security Officer of the Counterparty. Long-standing owner / admin on the Counterparty’s GitHub organizations: historically **satschel/***, and currently **liquidityio/*** — the latter being the org under which the rebuild integrating unlicensed Lux modules is being performed. The CSO role at any regulated digital-asset venue carries the duty to oversee software-supply-chain integrity, including open-source-license review and third-party-IP compliance. As admin on the Counterparty’s own repositories, Mr. Sonsurkar had on-platform visibility into the imported Lux LICENSE files — LEL, LRL-PR, BSD-3, and others — as they

entered the Counterparty’s build tree. (For clarity: `luxfi/*` GitHub-org admin was held by Mr. Nandy and Mr. Kelling on the Lux side; Mr. Sonsurkar did not have, and was not required to have, admin access on Lux’s own repositories. His duty of OSS-license review attaches to imports into the Counterparty’s tree, where his admin access was present.)

Co-Plaintiffs / Co-Victims

This memorandum is prepared on behalf of **Lux Industries Inc.** as primary plaintiff and **Hanzo, Inc.** as **co-plaintiff / direct co-victim with independent and additive claims**:

- **Hanzo, Inc.** is owned by substantially overlapping investors / partners with Lux Industries Inc. Hanzo operates the AI and infrastructure stack that is contributed back to the Lux upstream as OSS, and is the **source-licensor of the Hanzo-origin portion of the work product (the cryptographic primitives library, the consensus-algorithm prototypes, the 2018 ESX-JV BD + ATS development, the Beyond Equity / Beyond Alpha precedent exchange tech, and the early DEX work)** that was licensed / assigned forward into the Lux line of vehicles for Lux’s use (see §IV.A(1B)). **No such license or assignment was ever made from Hanzo, Inc. to the Counterparty.** Hanzo therefore holds its own independent and additive claims against the Counterparty in respect of the Hanzo-source portion of the contributed work product, separately pleadable as Counts I (DTSA), II (Copyright), III (Patent), V (Aiding-and-Abetting), VI (Unjust Enrichment), VII (Conversion), and XII (Civil Conspiracy) in Hanzo, Inc.’s name. The Counterparty’s exposure on the Hanzo- source layer is independent of and additive to its exposure on the Lux-source layer; any partnership-or-settlement construction contemplates release of **both** Lux Industries Inc. and Hanzo, Inc. on final execution (§XI ¶21–22), or none. Hanzo’s forward operational requirements additionally include **permanent access to the Counterparty’s protocol and venue for AI / infra-platform purposes** (§XI ¶21).
- **Zoo (Zoo Labs Foundation)** likewise requires permanent access to the Counterparty’s protocol and venue for AI- research / decentralized-science purposes (see §XI ¶21).
- The Beyond Equity Markets and Beyond Alpha Ventures trading- platform monorepos (referenced in §IV.A(3) as the precedent Alpaca-integrated exchange tech) are Lux / Hanzo work product, not Counterparty work product; their inclusion in this memorandum as evidence of pre-Counterparty exchange-tech precedent runs to both Lux and Hanzo standing.
- The **2018 ESX JV BD + ATS development work product** (Hanzo, Inc. / Ryan Kavanaugh / Triller; Hanzo / Lux side received a 20% stake in “Entertainment Stock X”) is documented Hanzo-origin upstream that further supports Hanzo, Inc.’s direct co-plaintiff standing under Counts I–III, V–VII, and XII.

For purposes of the substantive counts and damages analysis in §V–§VI, “Lux” should be read to include Hanzo where the context requires (i.e. for the OSS contribution chain-of-title, for the Hanzo-source IP layer never licensed to the Counterparty, and for the permanent-access mandatory term in §XI ¶21).

Counterparty Entity Structure

For purposes of this memorandum, “the Counterparty” refers collectively to the operating entities under the current capital raise (a Delaware corporation parent and its US LLC operating affiliate,

transacting as Liquidity.io) unless context requires otherwise. The corporate structure should be confirmed during outside-counsel diligence.

Lead investor (current round). Forum Markets, a publicly- traded company (formerly Ethzilla; ticker **FRMM** (NASDAQ; CIK 0001690080; principal office Palm Beach, FL; SIC 6199)), continues in the lead-investor role from the prior round and has signalled continued leadership of the current \$75M raise at the \$1.1B valuation. Forum Markets' public- issuer status means any decision to anchor a material investment in the Counterparty is potentially material to Forum Markets' own shareholders, with attendant Form 8-K considerations.

III. Jurisdiction, Venue, and Applicable Law

Federal causes of action sound in: Defend Trade Secrets Act (DTSA), 18 U.S.C. §§1836–1839; Copyright Act, 17 U.S.C. §§101 *et seq.*; Patent Act, 35 U.S.C. §§271, 281, 284, 285; Lanham Act, 15 U.S.C. §1125(a); Securities Exchange Act of 1934, 15 U.S.C. §78j(b) (Rule 10b-5, 17 C.F.R. §240.10b-5). **State and common-law claims** sound in Delaware (breach of fiduciary duty, *Aronson v. Lewis*, 473 A.2d 805 (Del. 1984); aiding and abetting, *Malpiede v. Townson*, 780 A.2d 1075 (Del. 2001)), and California (the likely Counterparty principal place of business): common-law fraud, unjust enrichment, conversion, tortious interference with prospective economic advantage, and civil conspiracy.

Forum recommendation: Federal District Court for the Northern District of California for the principal action, with related Delaware Chancery proceeding for the corporate-governance counts. Personal jurisdiction over the individual Counterparty officers is established by their commercial activities directed at Lux (a Delaware corporation) and their California residence.

Litigation hold — Lux should issue a formal preservation notice to the Counterparty and to each named individual within 30 days of any term-sheet delivery. Failure of preservation supports an adverse-inference instruction at trial.

IV. Statement of Facts

The following allegations are stated on information and belief, based on the personal knowledge of the undersigned in his dual operational role, supplemented by Lux's internal records.

A. Background of Lux Industries Inc. and Its IP (2008–2026)

1. All IP at issue is owned by Lux Industries Inc. The IP was developed across a continuous corporate-identity lineage beginning in 2008, with successive assignment of work product forward through each corporate vehicle until consolidation at present-day Lux Industries Inc.:

- **2008+ — Monoid LLC:** Mr. Kelling's solo Advanced AI / ML / Blockchain research firm. Monoid carried the foundational cryptographic-primitives research, the early machine- learning systems work, and the earliest distributed-systems engineering work that subsequently fed forward into the consensus and matching-engine designs at issue. **[Outside counsel to enumerate additional earlier research vehicles in the 2008–2016 window during diligence.]**
- **2016 — Hanzo, Inc.** (Delaware corporation) incorporated, succeeding the prior **Crowdstart** venture operated by Mr. Kelling. The Lux work continued under Hanzo, Inc. as the research and

engineering vehicle for the next phase — the cryptographic primitives library, the consensus-algorithm prototypes, and early DEX work that became the Beyond Equity / Beyond Alpha precedent exchange tech (§IV.A(3)). **In 2018, Hanzo, Inc. pursued its own original BD + ATS licensure development through a joint venture (the “ESX JV”) with Ryan Kavanaugh and the Triller group; the Hanzo / Lux side received a 20% stake in the JV vehicle (“Entertainment Stock X”), and the BD + ATS development work product was assigned / licensed forward into the Lux line of vehicles. The broader research stack that subsequently became Lux-branded IP was likewise assigned forward into the Lux line; Hanzo, Inc. retains a residual licensor / co-source posture in respect of the contributed work, and Mr. Kelling is currently compensated in material part through Hanzo, Inc. consulting fees (see §IV.A(1B)–(1C) below).**

- **[period to be confirmed by counsel]** — IP and team transitioned to **Lux Partners Limited**, an Isle of Man (IOM) private company limited by shares, which carried the research and early-stage product development through the next phase, including the original DEX matching-engine and threshold-MPC work.
- **Currently** — IP and operating activity sit in **Lux Industries Inc.**, the present-day Delaware corporation that holds **all of the IP at issue** in this memorandum and is the operative Counterparty-facing licensor of the LEL- and LRL-PR-tier modules.

The corporate-identity chain is unbroken in respect of the underlying IP (Monoid LLC + earlier research vehicles (2008+) → Hanzo, Inc. (2016, prior Crowdstart venture) → Lux Partners Limited (IOM) → Lux Industries Inc. (currently)), and the operative Counterparty-facing licensor in 2026 is Lux Industries Inc. alone. **[Outside- counsel diligence to confirm: full enumeration of pre-Hanzo research vehicles; the assignment instruments between Monoid LLC, the additional pre-2016 vehicles, Hanzo, Inc., Lux Partners Limited (IOM), and Lux Industries Inc.; existence of those instruments establishes unbroken chain-of-title for the patent, copyright, and trade-secret counts.]**

1A. Mr. Kelling’s pre-Monoid precedent context. As supporting context for the chain-of-attribution and originality claims at §IV.A(4) (patents) and the damages analysis at §VI, Mr. Kelling’s professional history before Monoid LLC includes relevant founding- / officer-level experience in early digital securities and equity-financing technology: **Chief Technology Officer of Crowdfunder.com**, an early equity-financing platform; and prior work on **Ar.ca**, an early digital-securities platform / DEX. This pre-Monoid record establishes the technical depth and continuity that underwrites the originality and uninterrupted attribution arguments in §IV.A(3)–(4). *Counsel may include or omit at discretion based on diligence; it is preserved here for completeness.*

1B. The Hanzo-source-license layer, and why it does not extend to the Counterparty. A material portion of the work catalogued in §IV.A(3) was originally developed under Hanzo, Inc. (2016 onward) and was **licensed and / or assigned forward into Lux** (Lux Partners Limited (IOM), then Lux Industries Inc.) for Lux’s operative use. **No such license or assignment was ever made from Hanzo, Inc. to the Counterparty (Liquidity.io or any of its affiliates).** The Counterparty therefore has **two independent licensor lacunae** in respect of the modules at issue: it lacks the operative LEL v1.2 / LRL-PR v1.0 license from Lux Industries Inc. (the Lux-side gap that drives Counts I–III and VI–VII of this memorandum), **and** it lacks any upstream license from Hanzo, Inc. in respect of the Hanzo-origin portion of the contributed work product. Hanzo, Inc. is therefore named as a co-Plaintiff / co-victim at §II (Parties and Dramatis Personae) and the Counts that are pleadable on the Hanzo-source layer (specifically Counts I (DTSA), II (Copyright),

III (Patent), VI (Unjust Enrichment), and the conspiracy-and-aiding-and-abetting counts (V, XII)) run in Hanzo, Inc.'s name as well as Lux Industries Inc.'s, in addition to the unitary Lux counts. The Counterparty's exposure on the Hanzo-source layer is independent of and additive to its exposure on the Lux-source layer; the partnership construction in §§X–XI accordingly contemplates a release of **both** Lux Industries Inc. and Hanzo, Inc. on final execution (§XI ¶21–22), or none.

1C. Mr. Kelling's current Hanzo consulting fees as evidentiary fact. Mr. Kelling is presently compensated, in material part, through **Hanzo, Inc. consulting fees**. This arrangement is consistent with Mr. Kelling's continuing role as a contributor and intellectual-property source whose work has flowed forward into the Lux line of vehicles for Lux's use. **It is not consistent with — and provides no support for — any claim that Mr. Kelling carried a free-and-clear personal license from Hanzo, Inc. or from any predecessor entity into the Counterparty.** The modules and engineering output Mr. Kelling contributed into the Counterparty's build during October–December 2025 and onward remain subject to (i) the Lux licensor posture for the Lux-branded modules, and (ii) the residual Hanzo, Inc. source-licensor posture for the Hanzo-origin portion. This fact pattern materially strengthens the Unjust Enrichment (Count VI), Conversion (Count VII), and Aiding-and-Abetting (Count V) theories.

1D. The technology stack so developed includes consensus algorithms, post-quantum cryptographic primitives, threshold and MPC protocols, fully-homomorphic-encryption (FHE) precompile architecture, virtual-machine substrates, EVM derivatives, decentralized-exchange matching engines, and cross-chain messaging protocols.

2. Lux's direct development investment in this stack exceeds \$20 million in recorded cost from 2016 to date [**evidentiary source: sworn declarations of corporate-finance officers across Hanzo, Inc., Lux Partners Limited (IOM), and Lux Industries Inc., consolidating recorded personnel + contractor + infrastructure cost across the lineage period — to be prepared by counsel in advance of filing**]. Fully-loaded personnel cost over the same period is a multiple of that figure.

3. Chronological development milestones:

- **2016 – 2017:** Founding under Hanzo, Inc. Foundational consensus research; early Quasar prototypes; cryptographic primitives library; secp256k1, Ed25519, BLS12-381 reference implementations.

Hanzo, Inc. client / advisory portfolio in this period (IP-relevant to the present matter):

- **Unikrn / Unikoin Gold (2017)** — early regulated e-sports bookmaker; Hanzo client / technical advisor on the **2017 Unikoin Gold ICO**, which raised approximately **120,000 ETH**. Hanzo, Inc. contributed cryptographic, smart-contract, and infrastructure work that subsequently fed forward into the Lux line of vehicles.
- **Ikigai Fund** — Hanzo, Inc. client; technical contribution that subsequently fed forward into Lux.
- **Ar.ca** — early digital-securities platform / DEX; Hanzo, Inc. client and engineering counterparty; predecessor technical work that informs the originality and chain-of- attribution analysis for the present DEX / matching-engine modules. Mr. Kelling additionally held CTO-level operating involvement at Ar.ca (§IV.A(1A)).

The aggregate IP output of these engagements — the cryptographic primitives, the early smart-contract and digital-securities patterns, and the operational platform work — **was licensed / assigned forward into the Lux line of vehicles** on the same chain-of-attribution basis as the ESX JV work product (§IV.A(1)). Hanzo, Inc. retains the residual source-licensor posture in respect of those contributions, and **no parallel licence was granted from Hanzo, Inc. to the Counterparty** (§IV.A(1B)).

- **2018 – 2020:** Active development of the planet-scale DEX matching engine with 1ms finality. Initial EVM fork (LGPL). Threshold-MPC research. **Hanzo, Inc. pursued original BD + ATS licensure development through the *ESX JV* with Ryan Kavanaugh / Triller (Hanzo / Lux side received a 20% stake in “Entertainment Stock X”); the BD + ATS work product from that JV was assigned forward into the Lux line of vehicles.** CDAX Limited (IOM affiliate) operated under an **IOM Class 8 money-transmission licence** in the same period, providing the operational compliance footprint for the IOM operations.
- **2020 – 2023:** Quasar consensus stabilization; Pulsar and Magnetar Apache-2.0 threshold-PQ primitives; FHE prototypes.

Luxembourg digital-securities + tokenized-commodity platform (Creatrust fund-manager partnership): in the same period, Lux (and Lux Partners Limited as predecessor) built and operated a regulated digital-securities platform in **Luxembourg** (CSSF-supervised footprint, §VIII Historical native support) in partnership with **Creatrust** as the licensed Luxembourg fund-manager counterparty. The platform covers (i) tokenized commodities — **uranium, gold, and silver**; (ii) tokenized fund-of-funds (FoF) structures; and (iii) tokenized fund LP interests — all issued on Lux’s **ERC- 3643 (T-REX)**¹ compliant smart-contract suite. The **same exact ERC-3643-compliant contract layer** Lux developed and deployed for the Creatrust / Luxembourg digital-securities platform is the underlying contract stack now embedded in the Counterparty’s ATS build (§IV.C) — which is independently dispositive of the chain-of-attribution and unjust-enrichment counts because there is **no plausible independent-derivation defence** for the Counterparty’s smart-contract layer when the same contracts (down to function signatures and storage layout) are deployed and diligenceable on Luxembourg-CSSF supervised infrastructure that predates the Counterparty’s build.

- **2024:** Ringtail lattice-signature scheme; FIPS 203 / 204 / 205 verifier precompiles (ML-KEM, ML-DSA, SLH-DSA); P3Q protocol; LEL v1.2 and LRL-PR v1.0 formalized; Warp messaging protocol stabilization; FHE-based dark-pool matching.
- **2025:** LX Suite DEX precompile family (Pool / Oracle / Router / Hooks at LP-9010 onward) released following Uniswap v4’s hooks model; Corona Ring-LWE threshold scheme (successor to Ringtail); LP-4200 precompile address standardization; multi-arch GPU acceleration under the closed `~/work/luxcpp/*` tree.

October–December 2025 (critical Counterparty-enabling work): Mr. Kelling personally introduced **Alpaca** (the FINRA-registered broker-dealer that currently serves as the Counterparty’s clearing BD) to the Counterparty, and over October–December 2025 personally developed (and integrated, with the Lux engineering team’s substantial contribution) the exchange-side technology that allows the Counterparty to interoperate with Alpaca’s clear-

¹ERC-3643 / T-REX is the Tokeny-stewarded EVM standard for permissioned, regulator-friendly security tokens; the standard is the de facto EU-compliant security-token substrate.

ing, custody, and securities- movement APIs. **Precedent context:** the Alpaca-integrated exchange-tech stack Mr. Kelling brought to the Counterparty is a substantial reuse / port of the prior **Beyond Equity Markets** (~/work/beyondequity) and **Beyond Alpha Ventures** (~/work/beyondalpha) trading- platform / unified-financial-platform monorepos built by the Lux / Hanzo engineering team prior to the Counterparty engagement. The Counterparty's current Alpaca-integration layer is therefore not Counterparty-original work — it is a port of pre-existing Lux / Hanzo platform code, which materially extends the IP-attribution analysis and the unjust-enrichment count. **This work is foundational to the Counterparty's ATS having been able to launch at all in its current commercial form:** absent Alpaca clearing and the Lux-built Alpaca-integration layer, the Counterparty had no operational path to executing customer trades under a SEC-registered clearing arrangement. The relevant Alpaca-side diligence material (Live Broker-API partner forms, beneficial- ownership certifications, DD checklists signed by Counterparty officers) is preserved as evidentiary record at liquidity/exhibits/alpaca-liquidity-bd/ adjacent to this memorandum. Damages attribution and partnership equity-share both weight this Alpaca-integration contribution heavily; absence of any contemporaneous compensating consideration to Lux for this work further supports the unjust-enrichment and unjust-licensing positions (Counts II, VI, IX).

- **2026:** Brand-neutral cleanup; relicensing of Pulsar / Magnetar under Apache-2.0; ongoing institutional partnership development.

4. Patent portfolio — bifurcated between Lux Industries Inc. and Hanzo, Inc. licensors.

Lux Industries Inc. portfolio — holds or has filed for patents on, without limitation: Quasar BFT consensus; sampler-bias correction; Ringtail / Corona lattice threshold schemes; Unified VM Execution Interface with Pluggable Backends; Post-Quantum State Verification with Verkle Proofs; FHE precompile architecture; GPU-accelerated DEX matching engine; AI-optimized bytecode execution. Source whitepapers and patent-grade specifications preserved at ~/work/lux/papers/ (LP-* series, including lp-020-quasar-consensus, lp-073-pulsar, lp-137-gpu-crypto-stack, lp-305-quantum-secure- assets, lp-307-quantum-consensus-protocol, lp-308-quantum-amm, and the lux-ats-architecture + gpu-evm-whitepaper families).

Hanzo, Inc. portfolio — holds or has filed for patents on, without limitation: AI / agent runtime architecture (Hanzo ACI, Agent SDK, AGI platform); model-context-protocol (MCP) infrastructure; AI-chain (ACI) cryptographic primitives bridging inference and consensus; Crowdstart-vintage commerce / equity- financing patterns; defense-grade AI / cross-domain / assured- networking primitives. Source whitepapers preserved at ~/work/hanzo/papers/ (including the defense/ subtree, hanzo-aci, hanzo-agent- *, hanzo-agi-platform, hanzo-ai-chain, hanzo-consensus-ai, hanzo-cloud-infrastructure, hanzo-blockchain-platform, and the jin/papers subtree).

Private GPU / C++ code layer. A material portion of the GPU-acceleration code (the ~/work/luxcpp/ **private tree** and the GPU-EVM kernels) is maintained as **private, non-open-source** source code by Lux and is patent-bearing. This layer is independently a trade-secret (Count I — DTSA) and is patent-protected (Count III — Patent). The Counterparty's commercial deployment of any binary or shared- object built against this layer is independently actionable on both counts.

Cross-portfolio standing. Because patents in the *Lux* portfolio and patents in the *Hanzo, Inc.* portfolio are both implicated by the Counterparty's current stack (the Lux-branded modules touching the Lux portfolio; the Hanzo-AI / ACI / agent / defense substrate touching the Hanzo portfolio, including via the Lux → Hanzo upstream OSS contribution chain), **Count III (Patent Infringement)**

is pleaded in the names of both Lux Industries Inc. and Hanzo, Inc. as co-plaintiffs.

Catalog summary (full diligence-ready table at `liquidity/review/patent_catalog_2026_05_25.md`, adjacent to this memorandum):

- **Eighty-seven (87) patent-bearing inventions** cataloged in total.
- Lux Industries Inc. exclusive: **47**; Hanzo, Inc. exclusive: **32**; joint Lux + Hanzo: **8**.
- Patent posture: **11 filed**; 51 filable; 21 defensive publications; 7 trade-secret records.
- Counterparty **probable read-ons: 23 inventions** (across Quasar 3.0 consensus, GPU-accelerated EVM (`cevm`), Warp 2.0 cross-chain finality, Block-STM parallel execution, Lightspeed DEX matching engine, Quantum AMM + perpetuals, Active Semantic Optimization, ML-DISA-65 / Groth16 validator binding, Lux Bridge protocol, and related primitives).
- **Top ten inventions by damages exposure** ($3.1\times$ to $5.2\times$ exposure multipliers on a comparable-transaction basis) sit at the core of the Counterparty’s stated “1 ms finality” and “GPU-accelerated matching” architecture; the Quasar 3.0 consensus and the Lightspeed DEX matching engine are the primary litigation anchors.
- **Estimated aggregate Count III damages exposure** (reasonable royalty floor under 35 U.S.C. §284, before willful- infringement trebling and before any DTSA / Copyright add-ons): **\$180M – \$420M, base case \$280M.**

[Outside counsel to verify USPTO filing status and PCT extensions of each enumerated invention; perfect any provisional filings to non-provisional within the standard 12-month window for any invention in active commercial use; coordinate Copyright Office registrations for Count II availability of statutory damages.]

B. The Lux Licensing Tier Structure

5. Lux publishes its software under a three-tier strategy:

6. **Tier 1 (BSD-3-Clause):** commodity infrastructure, free commercial use. `luxfi/node`, `luxfi/sdk`, `luxfi/cli`, `luxfi/api`, `luxfi/codec`, `luxfi/database`, `luxfi/p2p`, and approximately twenty- five (25) other modules.

7. **Tier 2 (LEL v1.2):** source-available, patent- protected. Per LEL §§1 and 3, commercial rights granted only to “Authorized Networks” — the Lux Primary Network (NetworkID 1, ChainID 96369), Lux testnets and devnets, and “L1/L2/L3 chains descending from the Lux Primary Network.”

8. Per LEL §4, forks and competing networks prohibited. Per LEL §5, all patent rights reserved. Per LEL §8, license terminates automatically on breach. Modules: `luxfi/consensus`, `luxfi/crypto`, `luxfi/mpc`, `luxfi/threshold`, `luxfi/ringtail`, `luxfi/corona`, `luxfi/fhe`, `luxfi/precompile`, `luxfi/captable`, `luxfi/broker`, `luxfi/treasury`, `luxfi/transfer`, `luxfi/compliance`, `luxfi/hsm`, `luxfi/aml`, `luxfi/maker`, `luxfi/warp`, among others.

9. **Tier 3 (LRL-PR v1.0):** research-only license. Section 3 expressly prohibits commercial use; Section 4 reserves all patent claims. Modules: `luxfi/vm`, `luxfi/staking`, `luxfi/sampler`, `luxfi/dex` (Go bindings prior to its 2026 move to fully proprietary).

10. **Tier 4 (Closed Proprietary):** the `~/work/luxcpp/*` tree (CUDA / Metal / WebGPU /

FPGA / lattice GPU accelerators, the C++ DEX matching engine). Not currently in the Counterparty's production graph; access requires a separately negotiated Private-Moat Addendum.

C. The Counterparty's Commercial Use of Lux IP

11. The Counterparty operates a registered Alternative Trading System, broker-dealer, and transfer-agent business in the United States, commercially deploying the validator binary `1qd` (built from `liquidityio/node`) on three live blockchain networks (Devnet ChainID 8675312, Testnet ChainID 8675310, Mainnet ChainID 8675309), each of which incorporates dozens of Lux modules.

12. The Counterparty further deploys the Liquidity EVM, ATS, BD, and TA backend services, white-label tenants (MLC, VCC, TZERO, ReTokens, and others), and approximately 12,796 SecurityTokens on Devnet alone running on top of Lux precompiles.

13. The companion audit (referenced internally as the license inventory) catalogues every Lux module directly imported. **Twenty-three (23)** directly-imported modules carry LEL or LRL-PR tier licenses requiring an executed commercial license (verified count: **19 LEL + 4 LRL-PR**, enumerated against `go.mod` files across the Counterparty tree).

14. No commercial license has ever been executed between Lux Industries Inc. and the Counterparty. Outside counsel should verify the non-existence of any such instrument through document discovery.

15. The Liquid chains DO activate the Lux Warp messaging precompile at the canonical address `0x0200000000000000000000000000000000000005`, as verifiable in the Counterparty's own `configs/network/chains/C.json` and `cchain-genesis.json` files: `warpConfig: {blockTimestamp: 0, ...}` activates Warp at genesis, and the precompile allocation at `0x02000...05` carries `code: "0x01"`. **Warp activation strengthens, rather than weakens, the LEL/LRL-PR licensing claim: the Liquid chains are not merely consuming Lux modules in source form; they are operating as a derivative chain consuming the Lux interoperability plane in production. **Under the LEL Authorized-Network definition (LEL v1.2 §1, §3, §5), the operative test is the *Descending-Chain* criterion — whether the chain descends from Lux consensus, retains the Lux-defined precompile namespace, and remains within the Lux interoperability domain. The Liquid chains satisfy the Descending-Chain criterion in code and therefore fall within the LEL grant's commercial-use trigger, which requires an executed commercial license that has never been executed (see 14.).****

D. The Counterparty Officer Team and Knowledge

16. Mr. Eric Choi (CEO) holds ultimate operational responsibility for technology licensing and IP compliance. His signature appears (or will appear) on the subscription documents for the current \$75M / \$1.1B-valuation raise; representations therein sound in his name. As of the date of this memorandum the round has not closed; the duty of disclosure is therefore current and ongoing. **Mitigating context:** on information and belief, Mr. Choi was **not informed** of the LEL / LRL-PR licensing mismatch by Mr. Sonsurkar (CSO) or Mr. Trombley (CSO/Strategy) despite the affirmative duties of those officers to flag it. Mr. Choi's culpability is accordingly mitigated from knowing fraud toward an officer-level disclosure failure absent personal knowledge, which materially shifts the Count IX (10b-5) and Count XI (common-law fraud) analysis as to him.

17. Mr. Austin Trombley (CSO, prev. Acting CEO) — discussed in §II above, including the Franklin Templeton litigation history and the pre-Kelling ATS technical state.

18. Mr. Coleman Church (BD CEO) — FINRA-regulated responsibility for the BD’s technology integrity; representations on FINRA filings sound in his name.

19. Mr. Jayant Sonsurkar (CSO) — Long-standing owner / admin on the Counterparty’s GitHub organizations `satschel/*` (historic) and `liquidityio/*` (current rebuild org), giving him direct, contemporaneous, ongoing visibility into every Lux LICENSE file as those files entered the Counterparty’s build tree on import — the LEL, LRL-PR, BSD-3, GPL, LGPL, BUSL, MIT, and Apache markers travel with their source files into any tree that consumes them. `luxfi/*` GitHub-org admin was held by Lux’s own engineering team (Mr. Nandy, Mr. Kelling) on the upstream side; Mr. Sonsurkar did not have admin on `luxfi/*` itself, nor was he required to — his software-supply-chain integrity duty attaches to imports into the Counterparty’s own tree, where his admin access was present.

20. The CSO role at a regulated digital-asset venue carrying ATS / BD / TA registrations includes (without limitation) software-supply-chain integrity oversight, identification of LEL / LRL-PR / GPL / LGPL restricted licenses in deployed dependencies, escalation of licensing mismatches to the CEO / BD CEO / GC, and implementation of remediation.

21. On information and belief, Mr. Sonsurkar did not, at any time prior to the elevation of this matter by Mr. Nandy, raise the LEL / LRL-PR mismatch internally, request a commercial license, include the unlicensed-IP status in regulatory compliance reviews, or flag the issue during current-raise diligence. By contrast, Mr. Nandy (VPE at the Counterparty, concurrently CTO at Lux) did identify the mismatch through his own dual-role visibility and elevated it upstream. The omission by Mr. Sonsurkar — the officer to whom that identification properly belonged — is the central evidentiary fact supporting Count IV (breach of fiduciary duty) and Count V (aiding and abetting).

22. Messrs. Choi, Trombley, and Church each had operational or strategic visibility into the Counterparty’s technology stack during the relevant period; the precise scope of each officer’s knowledge will require diligence development.

E. The Position of Mr. Zach Kelling

23. Mr. Kelling is the Founder of both Lux Industries Inc. and the Counterparty. He **has departed Lux Industries Inc.** and is **currently Chief Technology Officer of the Counterparty (Liquidity.io)**. His operational authority at the Counterparty covers the platform rebuild and ongoing technical posture. The Counterparty’s commercial decisions and investor-facing representations remain with Mr. Choi (CEO) and Mr. Trombley (CSO).

24. As a former Lux Founder/officer/cryptographer who departed Lux Industries Inc. and took the Counterparty CTO role during which the Lux-origin IP rebuild occurred, Mr. Kelling is a **prospective named defendant** in the available causes of action — specifically Count I (Misappropriation of Trade Secrets, DTSA), Count II (Copyright Infringement), Count III (Patent Infringement), Count IV (Breach of Fiduciary Duty owed to Lux as former Lux fiduciary), Count VI (Unjust Enrichment), Count VII (Conversion), Count IX (Rule 10b-5 Securities Fraud, by virtue of his current CTO role during the active \$75M / \$1.1B-valuation raise), Count XI (Common-Law Fraud), and Count XII (Civil Conspiracy). Mr. Kelling’s Counterparty equity position is to be confirmed in diligence; his former-Lux minority-shareholder position is documented.

25. **Key-person retention — mandatory in any partnership outcome.** Despite Mr. Kelling’s prospective defendant posture in litigation, the partnership outcome (§X) requires him to **remain in place as Counterparty CTO post-settlement**. Lux’s equity stake in the Counterparty is

rendered substantially worthless without Mr. Kelling's continued technical leadership — the current Counterparty operational team, absent Mr. Kelling, is not viewed as competent to maintain or extend the rebuilt platform that underpins the Counterparty's enterprise value. The partnership deal therefore pairs:

- **Mutual general releases of Mr. Kelling individually** from all Counts in this memorandum (conditional on full execution of the partnership agreement and ongoing performance);
- **A multi-year key-person retention commitment** from the Counterparty to retain Mr. Kelling as CTO with defined good-leaver / bad-leaver triggers, garden-leave constraints, and an equity clawback or compensation structure tied to the partnership equity value;
- **Carve-outs** of Mr. Kelling from any litigation Lux brings against other Counterparty officers (Choi / Trombley / Church / Sonsurkar) if partnership outcome is achieved.

Without partnership, Mr. Kelling stands as a co-defendant on the Counts above and the releases are not available. **See §X for the partnership tier structure and §XI for the full mandatory-terms list including key-person retention.**

F. Pre-Kelling Technical State and the ACH Exploit

26. In the period immediately preceding Mr. Kelling's arrival as active Chief Technology Officer, the Counterparty's ATS platform was materially non-functional as a regulated alternative trading system. Specifically, on information and belief:

- The ATS **did not implement working time-price priority order matching** — the foundational ordering principle of every regulated US alternative trading system, required by SEC Reg ATS for any venue purporting to operate as a securities trading facility.
- No functioning ATS pipeline existed end-to-end.
- **One day prior to Mr. Kelling's arrival as active CTO, the platform suffered an ACH-funding exploit that resulted in the theft of approximately \$40,000** via a trivial attack vector that should have been remediated by even baseline security review. The exploit evidences both the security posture of the platform under the pre-Kelling officer team and the absence of competent CSO oversight.

27. Upon his arrival, Mr. Kelling and his team (substantially overlapping with the Lux engineering team) rebuilt the ATS / BD / TA stack from the ground up, importing the Lux modules described in §IV(C) as the foundation. **In substantial part, the technology that supports the current \$75M / \$1.1B-valuation raise was the Lux IP that Mr. Kelling brought across, integrated with the regulatory wrapper.**

28. The factual proposition that the Counterparty's enterprise value as represented to investors is materially attributable to Lux-origin technology contributed by Mr. Kelling and the broader Lux team is supported by the contemporaneous technical record. This context bears on damages attribution (§VI), Mr. Trombley's credibility as a technology witness, the breach-of-fiduciary-duty count, and the protective treatment of Mr. Kelling (§XII).

G. The Current \$75M / \$1.1B-Valuation Capital Raise (Forum Markets, Lead)

29. The Counterparty is **currently undertaking a capital raise of approximately \$75 million at a \$1.1 billion valuation**. The round is led by **Forum Markets** (a publicly-traded company, formerly Ethzilla; ticker **FRMM** (NASDAQ; CIK 0001690080; principal office Palm Beach, FL; SIC 6199)), continuing in the lead-investor role from the prior round. As of the date of this memorandum, the Counterparty has **verbal commitments covering at least approximately half of the \$75 million round** (i.e. approximately \$37.5M soft-circled). **The round has not closed**. Funds have not been wired against current-round subscriptions; subscribers have not been issued securities in respect of the current round; the Counterparty's current and ongoing duty of disclosure to those subscribers remains live until closing. The subscription agreements, PPM, and side letters require outside- counsel review for representations regarding (a) Counterparty ownership of the underlying technology stack, (b) licensing status of all third-party IP, (c) related-party transactions, (d) the CSO's and BD CEO's compliance attestations, and (e) technology disclosures in the risk factors section.

30. To the extent the current-raise subscription documents represent or omit material facts regarding the Counterparty's ownership or licensed access to Lux IP, those representations are independently actionable under Section 10(b) of the Securities Exchange Act and Rule 10b-5. See Count IX, *infra*.

V. Causes of Action Available to Lux Industries Inc.

Count I — Misappropriation of Trade Secrets (DTSA, 18 U.S.C. §1836) — Bifurcated Standing (Lux + Hanzo)

Lux Industries Inc. as plaintiff — Lux's unpublished optimizations, integration know-how, architectural documents, deployment recipes, MPC operational procedures, sampler-bias remediation, and **pre-publication private-repository code including the `~/work/luxcpp/ GPU layer`** qualify as trade secrets.

Hanzo, Inc. as co-plaintiff — Hanzo's unpublished AI / agent runtime know-how, the ACI cryptographic primitives, the defense-grade primitives, the Crowdstart-vintage commerce / equity- financing patterns, and the residual unpublished portions of the upstream OSS contribution chain that flowed into the Lux modules likewise qualify as Hanzo trade secrets; Hanzo's standing on Count I is independent of and additive to Lux's.

Both Mr. Trombley AND Mr. Sonsurkar hold continuing owner / admin access to the Counterparty's GitHub organizations (`satschel/*` and `liquidityio/*`), giving them **100% oversight and overview of every repository, every LICENSE file as those files entered the Counterparty's build tree, every commit, and every pull-request discussion**. Their continuing repository access without an executed Lux *or* Hanzo commercial licence, coupled with the licensing-mismatch silence on both sides of their respective duties (CSO software- supply-chain integrity / OSS-licence review for Mr. Sonsurkar; CSO/Strategy + prior Acting-CEO investor-representation responsibility for Mr. Trombley), supports misappropriation by improper means.

Remedies (joint): Injunction (18 U.S.C. §1836(b)(3)(A)); actual damages and unjust enrichment (§(B)); reasonable royalty in lieu (*id.*); exemplary damages up to 2× for willful and malicious misappropriation (§(C)); attorneys' fees (§(D)); seizure if record warrants (§1836(b)(2)). Damages run

separately to each co-plaintiff in respect of the trade secrets in its own record that are misappropriated.

Count II — Copyright Infringement (17 U.S.C. §501) — Bifurcated Standing (Lux + Hanzo)

Lux Industries Inc. as plaintiff — Lux owns copyright in every source file under `luxfi/*` (public open-source modules + the private `~/work/luxcpp/ GPU layer`). Copyright notices are present in source headers.

Hanzo, Inc. as co-plaintiff — Hanzo, Inc. owns copyright in every source file under `hanzoai/*` (and predecessor / sibling orgs reflecting earlier Hanzo client work including `ar.ca`, the `unikrn / unikoin-` gold engagement record, and the `crowdstart` commerce work product) to the extent any such source file or derivative work was reused or ported into the Counterparty's build tree.

The Counterparty has copied by direct import in `go.mod`, container-image embedding, binary distribution to validator nodes, and on-chain deployment of derivative SecurityTokens. Copying that reads on a `luxfi/*` header runs to Lux Industries Inc.; copying that reads on a `hanzoai/*` (or predecessor) header runs to Hanzo, Inc.

Remedies (joint): Actual damages and infringer's profits, §504(b); or statutory damages up to \$150,000 per work willfully infringed, §504(c)(2); injunction (§502); impoundment (§503); attorneys' fees and costs (§505). Statutory-damages aggregation runs separately to each co-plaintiff in respect of its own registered or registrable works. **[Outside counsel: confirm Copyright Office registration status of each module under both `luxfi` and `hanzoai` header; perfect any unregistered works prior to filing to preserve statutory-damages availability.]**

Count III — Patent Infringement (35 U.S.C. §271) — Bifurcated Standing (Lux Industries Inc. + Hanzo, Inc.)

Lux Industries Inc. as plaintiff — Lux holds or has filed for patents covering the inventions enumerated in §IV(A)(4) (Lux portfolio). The Counterparty's deployment of Quasar consensus, Ringtail / Corona thresholds, the sampler-bias correction, the Liquid EVM substrate, the FHE precompile activations, the DEX matching engine, and any binary built against the private `~/work/luxcpp/ GPU layer` each implicate one or more patent claims in the Lux portfolio.

Hanzo, Inc. as co-plaintiff — Hanzo, Inc. holds or has filed for patents covering the AI / agent runtime, MCP, ACI, Crowdstart-vintage commerce, and defense-grade AI / cross-domain / assured-networking primitives enumerated in §IV(A)(4) (Hanzo portfolio). To the extent the Counterparty's stack embeds or reads on these primitives (whether directly or via Lux upstream OSS contribution), Hanzo, Inc. holds independent and additive standing to assert Count III against the Counterparty.

Remedies (joint): Reasonable royalty floor (35 U.S.C. §284); lost profits where established; injunction (§283); treble damages for willfulness (§284); attorneys' fees in exceptional cases (§285). Damages run separately to each co-plaintiff in respect of the patents in its own portfolio that are infringed.

Count IV — Breach of Fiduciary Duty (Del., against Jayant Sonsurkar individually)

As a corporate officer, defendant owes the corporation duties of care and loyalty. Mr. Sonsurkar's failure to identify and flag the LEL / LRL-PR licensing mismatch throughout his tenure is at minimum gross negligence in the CSO duty of care; if the record supports knowledge or willful blindness, it crosses into duty-of-loyalty breach.

Count V — Aiding and Abetting Breach (Del., against Messrs. Choi, Trombley, Church)

Per *Malpiede v. Townson*, 780 A.2d 1075 (Del. 2001): (i) a fiduciary relationship; (ii) breach; (iii) knowing participation by the non-fiduciary; (iv) damages. To the extent the named officers had actual or constructive knowledge of the mismatch and continued to direct commercial operations, they aided and abetted Mr. Sonsurkar's breach.

Count VI — Unjust Enrichment — Bifurcated Standing (Lux + Hanzo)

The Counterparty's commercial deployment of Lux *and* Hanzo IP without an executed license enriches the Counterparty at the expense of each upstream licensor; the LEL and LRL-PR (Lux) and the residual Hanzo source-licence posture (§IV.A(1B)) each expressly negate any justification. Standing runs separately to each co-plaintiff in respect of its own contributed work product.

Count VII — Conversion — Bifurcated Standing (Lux + Hanzo)

The Counterparty's deployment of LRL-PR-protected patent-bearing Lux modules constitutes intentional interference with Lux's exclusive rights; deployment of the Hanzo-source AI / agent / ACI primitives constitutes parallel interference with Hanzo's exclusive rights. Each interference is independently actionable and runs to its respective licensor.

Count VIII — Tortious Interference with Prospective Economic Advantage

The Counterparty's white-label tenants (MLC, VCC, TZERO, ReTokens) constitute particular instances of interference with Lux's commercial-license pipeline with third-party ecosystem partners.

Count IX — Securities Fraud (Rule 10b-5)

To the extent the current \$75M / \$1.1B-valuation raise is supported by representations or omissions regarding the Counterparty's IP ownership / licensing status, elements of Rule 10b-5 are met. **Lux's primary utility on this count is collateral leverage rather than direct recovery; standing analysis is for outside counsel.**

Count X — Lanham Act False Advertising (15 U.S.C. §1125(a))

The Counterparty's marketing materials describe its technology stack (post-quantum cryptography, FHE-based dark-pool matching, MPC custody, multi-chain architecture, Quasar-derived consensus) in terms that ascribe ownership or proprietary control without disclosing the Lux origin or the conditional / unlicensed nature of the access.

Count XI — Common-Law Fraud

To the extent Counterparty officers made representations to Lux personnel that licensing arrangements would be formalized in due course, and Lux relied to its detriment by continuing engineering contribution without an executed agreement, common-law fraud lies.

Count XII — Civil Conspiracy

The Counterparty officer team's concerted commercial deployment of unlicensed Lux IP and capital-raising on the back of that deployment supports civil-conspiracy joint and several liability.

VI. Damages Analysis**Small Scenario (Path A — Negligent Posture)**

Back-license value (12 yr at \$2.5M/yr base)	\$30M
Pre-judgment interest (5%, simple)	\$10M
Royalty component (1.5% of estimated GR)	\$10M
Forward-looking license value (NPV)	\$30M
Subtotal — Small Scenario	\$80M

Medium Scenario (Path B — Mixed-Knowledge, Contested)

Small-Scenario base	\$80M
DTSA 2× exemplary on willful misappropriation	+\$30M
Patent willfulness component (selective trebling)	+\$40M
Disgorgement on current-raise attribution (5% of enterprise value, TODO: rebase per counsel)	+\$55M
Attorneys' fees and costs	+\$10M
Subtotal — Medium Scenario	\$215M

Big Scenario (Path B Fully Litigated, Adverse Officer Record)

Base + interest + royalty	\$80M
DTSA full 2× exemplary	+\$80M
Patent full 3× trebling on willfulness	+\$120M
Disgorgement on current-raise attribution (15–20% of enterprise value, TODO: rebase per counsel)	+\$200M
Lanham Act + Lanham fees	+\$10M
Securities fraud (if Lux can establish standing)	+\$50M
Attorneys' fees, costs, pre-judgment interest	+\$50M
Subtotal — Big Scenario	\$590M

Opportunity-Cost Damages: 90-Day Bandwidth Diversion

The damages framework above does not yet capture a distinct and significant category of harm: the opportunity cost to Lux of having Mr. Kelling's — and the broader Lux

engineering team's — bandwidth diverted to the Counterparty's platform and to a constellation of related ventures during a critical 90-day window. The git record is unambiguous.

Quantification (rolling 90-day window) — Counterparty primary codebase. Audit of the liquidityio/* GitHub organization for commits attributable to Mr. Kelling:

Repo	Zach commits	Total commits	Zach %
liquidityio/ats	806	1,426	57%
liquidityio/bd	388	751	52%
liquidityio/ta	208	364	57%
liquidityio/mpc	31	95	33%
liquidityio/evm	23	32	72%
liquidityio/fhe	10	14	71%
liquidityio/iam	27	29	93%
liquidityio/node	338	380	89%
liquidityio/operator	354	406	87%
liquidityio/universe	1,436	1,920	75%
liquidityio/contracts	72	176	41%
liquidityio/dex	24	32	75%
liquidityio/platform	36	51	71%
liquidityio/superadmin	243	338	72%
liquidityio/exchange	522	1,170	45%
liquidityio/console	28	38	74%
liquidityio/ui	27	143	19%
liquidityio/sdk	30	30	100%
liquidityio/sdk-go	9	9	100%
liquidityio/cli	114	117	97%
Subtotal (Counterparty primary, 90-day)	4,726	7,521	63%

Quantification — related-venture codebases. The audit extends across five additional venture trees on which Mr. Kelling's bandwidth was concurrently directed during the same 90-day window:

Venture tree	Zach commits	LOC (add+del)	Note
VCC (vcc/exchange + contracts + wallet)	143	3,038,420	100% new venture (white-)
MLC (mlc/exchange + contracts + wallet)	153	3,044,377	100% new venture (white-)
EquityTable (loan-funding-engine + app)	98	681,666	100% new venture
OnyxPlus (admin + verify + onyxd + onboarding)	188	57,379	Largely 100% new venture
LiquidityArchive (legacy + simplici-*)	683	9,541,566	Legacy / archive maintenance
Subtotal (related ventures, 90-day)	1,265	16,363,408	

Grand totals across all six venture trees (90-day window):

Total Zach commits (all 6 trees)	5,991
Total LOC churn (add + del across all 6 trees)	~93 million
LOC churn in Counterparty primary (top 8 repos)	~77 million
Distinct working days in ats/ alone	55 (six days/week)
Repos where Zach >70% of all commits	12 of 20 Counterparty repos
Repos where Zach = 100% of all commits	2 (sdk, sdk-go)

Three of the five related ventures (VCC, MLC, EquityTable) were authored entirely

from scratch during the 90-day window — representing approximately **6.8 million lines** of net new code across companies that the Counterparty offers to its tenants and partners. A fourth (OnyxPlus, the KYC/ID joint venture) was largely new in the same window. This is not maintenance contribution; this is greenfield construction of multiple companies.

Mr. Kelling’s contribution to Lux upstream in the same 90-day window (the foundational Lux IP that powers the Counterparty’s stack):

Lux repo	Zach commits	Total commits
luxfi/node	0	179
luxfi/consensus	0	104
luxfi/crypto	0	64
luxfi/evm	0	99
luxfi/vm	0	27
luxfi/precompile	0	98
Total (90-day window)	0	571

The contrast is decisive: 5,991 commits and ~93M LOC of Zach’s bandwidth into Counterparty-side ventures versus exactly zero commits to the upstream Lux repositories he is Founder and Chief Cryptographer of.

Value-creation rate translation. The Counterparty’s reported current-round valuation in the active raise is approximately \$1.1 billion. The platform supporting that raise was substantially built or rebuilt by Mr. Kelling during the 90-day window. The implied value-creation rate is approximately **\$1.1B / 90 days = \$12.2M per day** of diverted attention — and that figure captures only the Counterparty’s primary codebase; it does not yet include the enterprise value of the three additional 100%-new ventures (VCC, MLC, EquityTable) and the largely-new OnyxPlus JV. Counting those, the per-day value-creation rate of Mr. Kelling’s bandwidth during the 90-day window is materially higher than the \$12.2M figure above.

The opportunity cost to Lux — measured by what Lux’s own roadmap would have produced during the same window had Mr. Kelling’s attention been available to Lux upstream — **is bounded above by that rate and below by the proportional founder- share of Lux’s enterprise trajectory.**

Component	Range
Bandwidth diversion at 50% rate (90 days)	\$500M – \$600M
At 25% rate	\$250M – \$300M
At 10% rate (conservative)	\$100M – \$120M
Conservative opportunity-cost claim	\$100–300M

This component is independent of and additive to the IP damages above. **It is the single most striking quantitative fact in the memorandum** and should be lead-in evidence for Mr. Dupont in opening negotiation.

Damages Summary (Including Opportunity Cost)

Scenario	IP damages	+ Opp. Cost
Small (Path A, settlement context)	\$80M	\$180–380M
Medium (Path B, mixed-record litigation)	\$215M	\$315–515M
Big (Path B, full-record litigation w/ multipliers)	\$590M	\$690–890M

Partnership remedy (headline). The figures above are **litigation-cash damage ranges** — the floor of what Lux could recover in court. Lux’s actual ask is a **partnership agreement** structured as a **10%–50% equity issuance in the Counterparty** plus a cash component plus reciprocal commercial commitments. At the **partnership floor (10% equity), nominal value is \$120–135M** (\$110M equity at \$1.1B-valuation pricing + \$10–25M cash). Recommended (25%) is \$300–325M. Stretch (35%) is \$435–460M. Maximum (50%) is \$600–650M. **Each tier carries escalating governance rights** (board observer → seat → parity) and broader reciprocal commitments (LEL grant → BaaS access → joint GTM → international distribution + closed-source GPU access). The partnership remedy dominates the litigation-cash damages because (i) it regularises the IP relationship directly, (ii) preserves Counterparty operational continuity which is a stated objective of Mr. Kelling (Counterparty CTO) and Mr. Nandy (Lux CTO / Counterparty VPE), (iii) prices off the same \$1.1B valuation Forum Markets and current-round investors are pricing into, and (iv) creates ongoing economic alignment between Lux and the Counterparty for the forward enterprise value Mr. Nandy’s team will continue building. See §X for the Partnership Triangulation matrix.

VII. Trading-Fee Revenue Projections (User-Scale Analysis)

The following projects the Counterparty’s enterprise-value trajectory at four user-base milestones and computes Lux’s economic interest at each under a proposed 10% equity stake.

Annual Revenue, Enterprise Value, and Lux 10% Stake

Scenario	Users	Revenue	Enterprise Value	Lux 10%
<i>Conservative ARPU (\$50/yr)</i>				
Stage 1	100K	\$5M	\$50M (10×)	\$5M
Stage 2	1M	\$50M	\$500M (10×)	\$50M
Stage 3	10M	\$500M	\$5B (10×)	\$500M
Stage 4	100M	\$5B	\$50B (10×)	\$5B
<i>Base ARPU (\$100/yr)</i>				
Stage 1	100K	\$10M	\$150M (15×)	\$15M
Stage 2	1M	\$100M	\$1.5B (15×)	\$150M
Stage 3	10M	\$1B	\$15B (15×)	\$1.5B
Stage 4	100M	\$10B	\$150B (15×)	\$15B
<i>Premium ARPU (\$200/yr)</i>				
Stage 1	100K	\$20M	\$400M (20×)	\$40M
Stage 2	1M	\$200M	\$4B (20×)	\$400M
Stage 3	10M	\$2B	\$40B (20×)	\$4B
Stage 4	100M	\$20B	\$400B (20×)	\$40B

Lux’s 10% equity, retained through the growth curve, is worth between \$5M (Stage 1 Conservative) and \$40B (Stage 4 Premium). Mid-case Base ARPU at 100M users implies

\$15B of Lux equity value.

VIII. Strategic Partnership Upside

Lux brings substantial commercial value to a partnership-resolution of this dispute. This section enumerates what the deal team can offer as upside in any negotiation tier.

First Major Prospective Client: AvaTrade — \$70B/month Trading Volume

Lux has direct access to AvaTrade as a prospective first major client for the combined Lux + Counterparty venue. The trading flow contemplated by that engagement is on the order of **\$70 billion per month** routed through the joint architecture. AvaTrade is an established multi-asset broker with operations across multiple regulated jurisdictions; the framework underlying this prospective arrangement is documented separately at `~/work/lux/legal/partnerships/avatrade/Lux_AvaTrade_`. *The engagement is in advanced discussion under that framework and is not yet a closed, signed customer commitment; the access to AvaTrade is the Lux contribution, the venue and clearing stack is the Counterparty contribution, and the joint take-rate is captured together.*

Fee economics. The Counterparty's existing cost stack on the securities-routing side is approximately **8 basis points blended**, which is consumed internally between the Counterparty's venue layer and the Alpaca clearing/custody relationship (split approximately 4 bps / 4 bps, subject to volume-tier adjustment). **Lux's take-rate sits on top of that internal cost stack and is structurally flexible in the 25–50 bps range:**

- **Conservative** (25 bps Lux take) on \$70B/mo = \$175M/mo = **\$2.1B/yr** in fee revenue to Lux from this single relationship.
- **Midpoint** (37.5 bps Lux take) on \$70B/mo = \$262.5M/mo = **\$3.15B/yr.**
- **Aggressive** (50 bps Lux take) on \$70B/mo = \$350M/mo = **\$4.2B/yr.**
- Even a **5 bps “partnership uplift” share-back** above existing cost to each of Alpaca and the Counterparty (10 bps combined incremental) is materially accretive to their per-trade economics and preserves the bulk of the Lux take.

Customer-facing total fee at the conservative end (~33 bps) remains competitive with multi-asset broker venues globally. At Stage-2-plus volume routing, the AvaTrade contribution **exceeds the Counterparty's current consolidated revenue base by approximately an order of magnitude** and is, by itself, the strongest commercial argument for partnership over litigation.

Crypto-Exchange Conversion at 8 Basis Points (Floor)

A specific deliverable Lux brings under the partnership: **the ability to offer crypto-exchange conversion at 8 basis points** — materially undercutting the incumbent fee structure at major US crypto venues (Coinbase / Kraken / Gemini at 50–200 bps; Robinhood PFOF-equivalent at 35–85 bps; institutional desks at 5–15 bps floor but with size minimums). The 8-bps floor is enabled by Lux's direct exchange and liquidity infrastructure and is a competitive moat for the joint Lux + Counterparty venue.

OnyxPlus Access from Counterparty at Cost (Reciprocal Term)

The Counterparty’s OnyxPlus identity-verification product (a Telos JV in which the Counterparty holds the operating relationship) provides identity verification infrastructure operating at the Lux ecosystem’s internal cost basis — materially below merchant pricing for equivalent KYC/AML/document/biometric services. Under the SCLA, **Lux receives access to the Counterparty’s OnyxPlus identity-verification product at no charge or at the Counterparty’s documented direct cost (no markup, no commercial margin), not at market rates.** The annual savings depend on the Counterparty’s user-onboarding velocity but at Stage-2-plus scale run into the tens of millions per year.

Co-Branding for Lux-Network Chains

All Lux-network L1/L2/L3/L4 chains receive explicit co-branding on the Counterparty’s customer-facing surfaces: the Counterparty’s marketing, the venue’s chain selector, the SecurityToken issuance pages, and any institutional partnership collateral. This brand co-positioning is a non-negotiable term under the SCLA (§XI).

Real-World Asset Pipeline — Lux Brings Issuers; Counterparty Earns Listing + Trading Fees

Lux operates the infrastructure layer behind multi-billion-dollar RWA tokenization pipelines, and **brings the issuers.** The pipeline is not theoretical — Lux already operates a **regulated Luxembourg digital-securities platform in partnership with Creatrust (the licensed Luxembourg fund-manager counterparty)**, on the same ERC-3643 / T-REX smart-contract suite (§IV.A(3) 2020–2023 milestone) that underlies the Counterparty’s ATS contract layer. The Luxembourg / Creatrust footprint covers **tokenized commodities (uranium, gold, silver), tokenized fund-of-funds (FoF) structures, and tokenized fund LP interests**, and provides both the CSSF-supervised regulatory wrapper and the multi-year operational record that anchors Lux’s issuer-side credibility.

Under the SCLA, the Counterparty is the natural US-regulated listing destination for that flow and **captures the listing fees and the recurring trading-fee stream** on every asset that goes up. The economic shape of the relationship is: Lux = issuer origination, structuring, contract layer (ERC-3643 / T-REX), and token-side cryptographic and chain-side machinery; Counterparty = regulated ATS / BD / TA listing venue + Alpaca clearing + customer-facing distribution; **fees are split per the SCLA fee schedule** (Lux take + Counterparty take + Alpaca clearing cost). Asset classes in the pipeline:

- Tokenized US private-company equity and securities.
- Tokenized real estate / NNN-SPV-wrapped property exposures.
- **Tokenized commodities — uranium, gold, silver, FX, energy** (in production today on the Creatrust / Luxembourg platform; SCLA migrates / cross-lists into the Counterparty’s US-regulated venue).
- **Tokenized fund LP interests and fund-of-funds (FoF) structures** (in production today on the Creatrust / Luxembourg platform).
- Tokenized fixed-income.

- Tokenized music + IP royalty streams.
- Tokenized art and collectibles (>\$1M lots).
- **Substantial equity portfolio under tokenization mandate (up to \$1 trillion in nominal value)** spanning AI companies, resource / mining / energy companies, and other sector-diverse portfolio holdings being prepared for tokenized listing on the joint venue. **Every asset listed on the Counterparty’s venue generates (i) a one-time listing fee and (ii) a recurring trading-fee stream over the asset’s lifetime on-venue;** the size and composition of the pipeline therefore materially shifts the partnership’s forward enterprise-value trajectory and is a primary driver of the equity-share floor in §X.

Conservative estimate of incremental deal flow under the SCLA: \$5–15 billion in nominal asset value over the first 12–24 months post-close on the multi-asset RWA pipeline above, plus the phased onboarding of the \$1 trillion equity-portfolio mandate over the medium term (24–60 months) as listings clear regulatory and custodial preconditions. Recurring trading-fee revenue on the seasoned book scales linearly with on-venue turnover.

Reciprocal License to Counterparty ATS / BD / TA / Listed Securities — Lux’s Forward Venue Strategy

Lux Industries Inc. does not presently operate an alternative trading system or broker-dealer in the United States. However, Lux’s historical regulatory posture and its forward strategy include venue licenses in multiple jurisdictions, and the reciprocal license to the Counterparty’s ATS / BD / TA / listed-securities surfaces described in §XI(9) is intended to give Lux operational optionality across that footprint as it matures.

Historical native support: Lux (and Lux Partners Limited as predecessor) previously operated regulatory arrangements in the **Isle of Man (IOM)** (IOM Financial Services Authority supervision) and **Luxembourg** (Commission de Surveillance du Secteur Financier engagement). The IOM and Luxembourg footprints provided native EU-passportable and offshore-fund-eligible regulatory reach. Within the IOM footprint, the Lux family previously held, through **CDAX Limited** (an IOM company in the Lux-affiliated corporate group), an **IOM Class 8 money-transmission licence**² — an operational licensing posture that establishes both the technical-operations and the compliance-program track record relevant to the forward US (Reg ATS / BD) and UK (FCA) applications and that is independently diligenceable in the IOMFSA register. **Within the Luxembourg footprint**, Lux operates a regulated digital-securities platform in partnership with **Crestrust** (the licensed Luxembourg fund-manager counterparty), running Lux’s ERC-3643 / T-REX contract suite to issue tokenized commodities (uranium, gold, silver), tokenized fund-of-funds, and tokenized fund LP interests — the Luxembourg posture provides an independently diligenceable CSSF-adjacent regulated venue and a multi-year operational record for the same contract layer now embedded in the Counterparty’s ATS build.

Prior BD + ATS development — the 2018 ESX JV. Concurrent with the IOM Class-8 footprint, in 2018 **Hanzo, Inc. pursued original BD + ATS licensure development through a joint venture (the “ESX JV”) with Ryan Kavanaugh and the Triller group** (the social / short-form video group that subsequently matured into the public Triller listing). The economic

²IOM Financial Services Authority licensing class covering money transmission and ancillary payment services; see IOMFSA Regulated Activities Order.

outcome of the JV was that the Hanzo / Lux side received a **20% stake in “Entertainment Stock X”** (the JV vehicle), and the BD + ATS development work product generated within that JV was **assigned / licensed forward into the Lux line of vehicles** through the same corporate-identity chain documented at §IV.A(1). The ESX JV work product is therefore part of the upstream record that materially **predates** the Counterparty’s 2025–2026 build of its own ATS, supports the originality and chain-of-attribution claims at §IV.A(4) (patents) and Counts I–III, and forms part of the Hanzo, Inc. source-licensor layer described at §IV.A(1B) — which, again, was **never licensed to the Counterparty**.

Forward expansion: Lux is actively evaluating venue-license applications across the United States (Reg ATS), United Kingdom (FCA MTF or equivalent), European Union (MiCA-compliant Trading Platform), Singapore (MAS Capital Markets Services Licence + Recognised Market Operator), and the UAE (VARA). Each license, when obtained, becomes a venue at which Lux may operate the Counterparty-licensed ATS / BD / TA technology under §XI(9).

Cross-listing rights: Every security that lists on the Counterparty venue is available for cross-listing on any Lux-network or Lux-affiliate venue, and every security that lists on a Lux venue is available for cross-listing on the Counterparty’s venue.

Banking Licenses — SF Private Bank Partnership and Global Coverage

Lux holds a strategic partnership with SF Private Bank (“SFPB”) and, through SFPB and its corporate parent Supreme Fintech, gains access to the consolidated SFPB / Supreme Fintech licensed footprint across the populated continents. Adjacent partnerships with additional regulated financial institutions fill remaining geographies. The combined licensed coverage:

- **North America:** United States (federal banking partner, FDIC-insured rails); Canada (chartered banking partner, CDIC-insured).
- **Europe:** Sweden (Finansinspektionen-supervised, EU passport); historical IOM (Financial Services Authority); historical Luxembourg (CSSF); ongoing UK (FCA) and Switzerland (FINMA) expansion.
- **Asia-Pacific:** Singapore (MAS Capital Markets Services and banking); New Zealand (RBNZ-licensed deposit-taker); ongoing Hong Kong (SFC type-1 / type-7) and Japan (JVCEA) expansion.
- **Middle East:** UAE (VARA-supervised; DIFC / ADGM counterparty relationships).
- **Latin America:** regional partnerships under bilateral arrangements; specific licensing on a case-by-case basis.
- **Africa:** pilot arrangements; specific licensing on a case-by-case basis.

Services Lux can deliver under the SCLA include Banking-as-a-Service (BaaS), Payment Service Provider (PSP) rails, and **a custody solution that materially undercuts the Fireblocks and BitGo incumbents on price** (built on the same MPC infrastructure already in the Counterparty’s production).

Estimated independent cost for the Counterparty to acquire equivalent global license suite: \$57–140M in direct regulatory and capital expenditure, plus 5–7 years of timeline. See Schedule C.

International Distribution: M7 and 1B+ End-User Reach

Through Hanzo Inc., the broader corporate family, and the Lux ecosystem partnerships, the M7 global initiatives provide distribution access to more than 1 billion end-users across consumer, institutional, and retail channels. Post-SCLA, Lux’s international counterparty network operates as a Reg-S-compatible distribution channel for Counterparty-listed assets.

Hanzo Inc. Portfolio Precedents

Hanzo Inc., the venture studio under which the Lux relationship has historically operated, has applied the IP-in-kind founder-equity model successfully to multiple spawned unicorn-class companies:

- **Damon Motorcycles** — electric / smart-motorcycle company.
- **Triller** — short-form video / social platform.
- **Bellabeat** — women’s health wearables.
- **Unicorn Gold** — precious-metals tokenization platform.
- **Lux Network** — itself a Hanzo-spawn outcome.
- **Zoo Labs Foundation / Zoo Network** — decentralized AI research network.

Retained-parent-equity range across the portfolio: 10%–40%. The 10% ask in the Counterparty is at the bottom of that internal range.

Private IP & Patent Portfolio Access — \$1B+ Aggregate Valuation

In addition to the public-OSS-tier Lux modules and the LEL / LRL-PR source-available tiers (§IV.B), the Lux + Hanzo combined estate includes a **private, closed-source, patent-bearing IP portfolio** maintained outside the public repositories and not present in the `luxfi/*` or `hanzoai/*` GitHub organizations. Sources and scope (catalog being finalised in `liquidity/review/patent_catalog_2026_05_` adjacent to this memorandum):

- `~/work/luxcpp/` — closed-source GPU / C++ acceleration tree: GPU-accelerated DEX matching engine kernels; GPU-EVM execution kernels; GPU-accelerated cryptographic primitives (BLS multi-pairing, KZG, FFT, MSM); multi-arch (CUDA / Metal / ROCm / oneAPI) shim layer; AI-optimised bytecode execution. Patent-bearing on multiple claims.
- **Lux paper portfolio** (`~/work/lux/papers/`) — the LP-* whitepaper series and topical specifications underlying the patent filings enumerated at §IV.A(4) (Lux side), including `lp-020-quasar-consensus`, `lp-073-pulsar`, `lp-137-gpu-crypto-stack`, `lp-305` through `lp-309` quantum-stack family, the `lux-ats- architecture`, and the `gpu-evm-whitepaper`. Each paper constitutes a defensive-publication or filable-patent record.
- **Hanzo paper portfolio** (`~/work/hanzo/papers/`) — HIP-* and thematic papers including `hanzo-aci`, `hanzo-agent-grpo` / `hanzo-agent-sdk`, `hanzo-agi-platform`, `hanzo-ai-chain`, `hanzo-consensus-ai`, `hanzo-cloud- infrastructure`, `hanzo-blockchain-platform`, the `jin/` multimodal subtree, and the `defense/` subtree (assured-networking, cross-domain, countermeasures, edge-AI/ML). Each paper constitutes a defensive-publication or filable-patent record on the Hanzo side.

- **Adjacent private repositories** across ~/work/ bringing pre-existing trading- platform, exchange, AI / ML, and infrastructure work product into scope: ~/work/beyondequity, ~/work/beyondalpha, and other Lux / Hanzo internal monorepos. (Counsel: complete enumeration during diligence.)

Aggregate valuation: in excess of \$1 billion. The combined Lux + Hanzo private-IP and patents estate, valued on a comparable-transaction basis against precedent crypto and AI infrastructure IP transactions (and against the Counterparty’s own current \$1.1B raise valuation, which materially reflects the contribution of this IP via Mr. Kelling’s Q4-2025 rebuild), sits **north of \$1 billion in aggregate value** before any consideration of forward licensing or commercialisation revenue. That valuation is a function of (i) the number and breadth of patent-bearing inventions; (ii) the demonstrated reusability of the OSS-public layer to extract enterprise value (the Counterparty’s own raise being the immediate evidence); and (iii) the rarity of an integrated cryptographic + post-quantum + GPU + AI + exchange-engine portfolio at this depth.

SCLA grant scope. Under the SCLA, the Counterparty receives access to this private portfolio **at the Maximum (A+) tier only**, and at progressively more limited subsets at lower tiers (see §X feature-unlock table). At any tier, the SCLA grant runs to the Counterparty as licensee, subject to (a) non-sublicensability outside the Counterparty’s regulated venues, (b) the standard LEL / LRL-PR-style “Authorized Network” / “Descending Chain” criterion (§IV.B), and (c) the SCLA’s material-breach termination provisions. **The Hanzo, Inc. co-licensor consent (§XI ¶20, ¶22A) is required for the Counterparty to access the Hanzo-side portion of the portfolio.**

The \$100B–\$1T Target Institution

The combined Lux + Counterparty vision is the construction of a \$100-billion-to-\$1-trillion-class institution at the intersection of tokenized real-world assets, post-quantum-secure infrastructure, regulated US securities markets, and global distribution. The deal team should treat the negotiation not as a zero-sum settlement but as the founding-document of a partnership creating value materially in excess of the negotiating range.

IX. Partnership Now vs. Delay Analysis

Forces favoring delay: equity appreciation; royalty accrual; current-raise disgorgement expansion; pattern-of-conduct evidence.

Forces favoring settlement now: statutory damages caps don’t grow with delay (DTSA at 2×; patent at 3×); dilution risk on Lux’s equity component; operational risk (engineer-around); evidence preservation; DTSA 3-year limitations from discovery; Lux operating runway; negotiating leverage compounds short-term then evaporates if the Counterparty hits trouble; public visibility window.

Recommendation: Partner now, with strong protective terms — significant cash component, 10%+ equity with weighted- average broad-based anti-dilution, full ratchet on down-rounds, pro- rata participation, tag-along and co-sale rights, 1+1 mandatory board seats, 99-year exclusive USA-market license to the Counterparty (conditional), patent grant, AvaTrade routing, 8 bps conversion floor, OnyxPlus access from Counterparty at zero / documented cost, co-branding for Lux chains, and reciprocal permanent license to Counterparty ATS / BD / TA / listed- securities / cross-listings (§XI). **Statutory damages caps do not grow with delay; the equity does.** Locking equity now with strong anti-dilution captures the future-growth narrative without the operational risks of continued reliance on a deteriorating informal arrangement.

X. Partnership Triangulation (Equity-Share + Cash + Governance + Reciprocal Commitments)

Posture

This is a **partnership-structuring matrix, not a litigation- settlement matrix**. The deal team's job is to land a structure that both sides can live with — one that regularises the IP relationship, preserves the Counterparty's operational continuity (including Mr. Nandy's team's role), gives the Counterparty's current-round investors a clean cap table, and gives Lux Industries Inc. a substantive ongoing economic interest. Litigation (§V–VI) is the alternative posture if a partnership cannot be reached; the available exposure range is \$80M–\$890M cash and is documented as leverage, not as the operative ask.

Partnership Sizing Tiers (10%–50% equity range)

The deal range is **10% equity (minimum substantive partnership) to 50% equity (full equal-partner recapitalisation)**.

Cash and equity are inversely related to a constant total- value target. The reference point is the maximum-equity tier ($50\% \times \$1.1\text{B} = \550M , no cash required because equity alone satisfies the target). At lower equity percentages, **cash fills the gap**: $\text{cash} = (50\% - \text{agreed equity } \%) \times \1.1B . **Equity $\uparrow \Rightarrow$ cash $\downarrow$** . The Counterparty chooses where in the range to land based on its preference between existing-shareholder dilution (equity-heavy preserves cash) and cash outflow (cash-heavy preserves cap-table). Each tier additionally carries an escalating governance package and broader reciprocal commercial commitments (right column).

Tier	Equity	Cash	Total nominal	Governance + reciprocal
Maximum (M)	50%	\$0	\$550M (\$550M equity + \$0 cash)	Governance parity; equal-partner recap; full reciprocal license; international M7 distribution; closed-source luxcpp GPU access
Stretch (S)	35%	\$165M	\$550M (\$385M + \$165M)	Board seats (2); above + co-branding for Lux-network chains; AvaTrade flow routing; 8 bps crypto-conversion floor
Recommended (R)	25%	\$275M	\$550M (\$275M + \$275M)	Board seat (1); above + BaaS / SF Private Pay access; joint GTM commitments
Floor (F)	10%	\$440M	\$550M (\$110M + \$440M)	Board observer; LEL/LRL-PR perpetual license to Counterparty; reciprocal BD/ATS/TA access to Lux

The math: $\text{cash} = (50\% - \text{agreed equity } \%) \times \1.1B . So $50\% \text{ equity} \Rightarrow \0 cash ; $35\% \Rightarrow \$165\text{M}$; $25\% \Rightarrow \$275\text{M}$; $10\% \Rightarrow \$440\text{M}$. Every tier delivers the same **\$550M total nominal value** to Lux, which is the floor of Lux's claim (50% of enterprise value derived from Lux IP).

Practical constraint — cash sequencing. The Counterparty's post-current-raise cash position will not support large lump-sum cash payments at the lower-equity tiers in a single payment. The deal team should sequence the cash leg as a **multi-year schedule** (e.g. 30% at close, balance over

24–36 months from operating cash flow, secured by a security interest in Lux’s pledged equity stake or a default-to-additional-equity conversion clause). A multi-year cash schedule shifts the practical landing zone toward the higher-equity tiers — which is the natural commercial outcome and which is what Lux wants (more equity participation in long-term value).

Pricing Comp — Anchor on the Current Round

The Counterparty’s current \$75M raise is being subscribed at \$1.1B valuation (Forum Markets lead, \$37.5M soft-circled). Equity issued to Lux at the same valuation is priced on the same per-share basis as the new-money investors. This is the clean precedent transaction to anchor on — no separate valuation exercise required, no contested damages multiplier, no dispute over the basis of the equity value.

Feature-Unlock by Settlement Tier

Forward-value feature	S–	S	M	B	A+
LEL/LRL–PR perpetual license	✓	✓	✓	✓	✓
Patent grant on covered IP	✓	✓	✓	✓	✓
Maintenance & support of upstream OSS		✓	✓	✓	✓
99-year exclusive USA-market license		✓	✓	✓	✓
1 Lux + 1 independent director seat		✓	✓	✓	✓
RWA pipeline routing into Counterparty		✓	✓	✓	✓
AvaTrade flow routing into venue		✓	✓	✓	✓
8 bps crypto-conversion floor		✓	✓	✓	✓
OnyxPlus access from Counterparty at zero / documented cost		✓	✓	✓	✓
Co-branding for Lux-network chains		✓	✓	✓	✓
Reciprocal license to ATS/BD/TA			✓	✓	✓
SFPB banking-license access (BaaS, PSP)			✓	✓	✓
Custody undercutting Fireblocks			✓	✓	✓
International M7 distribution channels				✓	✓
Private Lux portfolio (luxcpp GPU + LP-* papers)				✓	✓
Private Hanzo portfolio (jin, defense, ACI, agent SDK papers)					✓
Full ~/work/* private estate (\$1B+ valuation)					✓
Joint go-to-market programs				✓	✓

Partnership vs. Litigation Triangulation

Path	Lux re
Partnership Maximum (M, 50%)	\$550M equity + \$0 cash + governance parity + full reci
Partnership Stretch (S, 35%)	\$385M equity + \$165M cash + 2 board seats + GTM + co-br
Partnership Recommended (R, 25%)	\$275M equity + \$275M cash + board seat + BaaS
Partnership Floor (F, 10%)	\$110M equity + \$440M cash (multi-year) + board observer + LEL
<i>Alternative path (litigation; documented as leverage, not as the operative ask):</i>	
Litigate: Small (Path A)	\$80
Litigate: Medium (Path B mixed)	\$215
Litigate: Big (Path B full)	\$590
Litigate + Opp. Cost (Big)	\$690–890

Every partnership tier strictly dominates litigation on a forward-value basis because (i) the equity component participates in the Counterparty’s upside, (ii) the reciprocal commercial commitments generate operating revenue for both sides, (iii) Mr. Nandy’s team remains in the build, and (iv) Lux’s IP is regularised cleanly under the LEL/LRL-PR perpetual license that ships with every tier. The litigation rows are documented for negotiating purposes and as the alternative posture if partnership terms cannot be agreed; they are not the operative ask.

XI. Mandatory Terms (Lux’s Floor in Any Partnership)

The following terms are non-negotiable. Any partnership outcome at any of the four tiers (Floor / Recommended / Stretch / Maximum) must include each of them.

0. Key-person retention — Mr. Zach Kelling remains as Counterparty Chief Technology Officer. Lux’s equity stake is rendered substantially worthless without Mr. Kelling’s continued technical leadership of the Counterparty platform; the current Counterparty operational team absent Mr. Kelling is not viewed as competent to maintain or extend the rebuilt stack that underpins the Counterparty’s enterprise value (see §IV.E ¶25 and §IV.F). The partnership therefore requires:

- Mr. Kelling’s continued tenure as Counterparty CTO for a minimum term of **four (4) years** from closing, with defined good-leaver / bad-leaver triggers and a 12-month garden-leave / non-compete on the Counterparty side if termination is without cause.
- Mutual general releases of Mr. Kelling individually from Counts I, II, III, IV, VI, VII, IX, XI, and XII of this memorandum, conditional on full execution of the partnership agreement and ongoing performance under it.
- Carve-out of Mr. Kelling from any litigation Lux may bring against other Counterparty officers (Choi / Trombley / Church / Sonsurkar) if partnership terms are achieved.
- Compensation structure (cash + Counterparty equity grants) benchmarked against the comparable CTO market and adjusted to align his continuing economic interest with both Lux (via partnership equity) and the Counterparty (via direct compensation).
- Bad-leaver clawback triggers tied to material breach of post-settlement obligations.

1. Board composition — expansion to a minimum of seven (7) directors total, with structured composition. The Counterparty’s existing three-director board failed to identify the licensing mismatch over the relevant period despite having a maintainer-access-empowered CSO whose duty included precisely that review. **The 3-person board structure is, in light of that**

record, an inadequate governance vehicle going forward, and the disinterested-investor case for board expansion is unambiguous.

The mandatory composition is:

- **Three (3) existing director seats retained** from the current board.
- **One (1) Lux director seat**, for Lux Industries Inc.'s direct representative (Mr. Pahlavi's designate).
- **Three (3) mutually-agreed independent director seats**, each nominee with no prior commercial or personal relationship to either Lux or the Counterparty, serving as the ongoing governance anchor of the SCLA.

Effective at closing. **No ownership-maintenance condition attaches to the Lux seat.** The structure ensures (a) no single faction (existing officers, Lux, or independents) holds majority voting power; (b) the independent block of three directors provides the disinterested-quorum threshold for any related-party transaction under Delaware General Corporation Law §144; and (c) the Counterparty's institutional governance moves from informal, founder-controlled posture to formal, multi-faction supervision appropriate to a venue valued in the \$1B+ range with FINRA / SEC exam exposure.

In the alternative (a 7-director soft floor that Lux will accept under appropriate compensating concessions), the composition may shift to: 2 existing seats retained + 2 Lux directors + 3 independent seats. This variant displaces one of the existing seats and strengthens Lux's voting position; Lux's deal team may invoke this alternative as leverage if the Counterparty resists the principal structure above.

2. 99-year exclusive USA-market license to the Counterparty for the LEL/LRL-PR portfolio in scope. Conditional on maintenance of all SCLA covenants. Material breach terminates exclusivity. Outside the USA, Lux retains the right to license to other parties.

3. Anti-dilution: weighted-average broad-based, with full ratchet on any down-round below the Lux issuance price.

4. Pro-rata participation right on all future Counterparty preferred financings.

5. Tag-along and co-sale rights on any founder transfer exceeding 1% of fully-diluted equity.

6. Information rights: quarterly and annual financials; all board materials and minutes; all share-issuance and option-grant resolutions; counsel and accountant access under NDA.

7. Patent grant on all current and future Lux patents covering the licensed IP. Worldwide; perpetual; royalty-free under the SCLA; expressly inclusive of patents not yet filed at SCLA execution.

8. Maintenance and support of upstream Lux OSS: continuing on the shared-engineering basis as today, no separate maintenance fee.

9. Reciprocity — Permanent license to Counterparty ATS, BD, TA, listed securities, and cross-listings. As symmetric consideration for Lux's perpetual IP license, the Counterparty grants Lux Industries Inc. (and any Lux-controlled affiliate or Lux-network L1/L2/L3/L4 chain maintaining Lux consensus and Warp interop):

- **Permanent, royalty-free, non-exclusive license to the Counterparty's ATS technology and operational know-how**, sufficient for Lux to operate (or sublicense to a Lux-

controlled regulated entity) an ATS in any jurisdiction where Lux holds or obtains the necessary venue license.

- **Permanent, royalty-free, non-exclusive license to the Counterparty's BD technology and know-how** on equivalent terms.
- **Permanent, royalty-free, non-exclusive license to the Counterparty's TA technology and know-how** on equivalent terms.
- **Reciprocal listing and cross-listing rights:** every security listed on the Counterparty venue is available for cross-listing on any Lux venue, and every security listed on a Lux venue is available for cross-listing on the Counterparty venue, on no-less-favorable terms.
- **Sublicensing rights:** Lux may sublicense to Lux- controlled jurisdictions and operating affiliates, including the IOM, Luxembourg, FCA, MAS, VARA, MiCA, and FINMA jurisdictional expansions described in §VIII.

The reciprocal grant runs 99 years, conditional on continued SCLA performance.

10. Co-branding for Lux-network chains. All Lux- network L1/L2/L3/L4 chains receive explicit co-branding on the Counterparty’s customer-facing surfaces (marketing, venue chain selector, SecurityToken issuance pages, and institutional partnership collateral).

11. OnyxPlus access from Counterparty at zero or documented cost. Lux receives access to the Counterparty's OnyxPlus KYC / AML / document / biometric identity-verification product (operated by the Counterparty as a Telos JV) at no charge or at the Counterparty's documented direct cost (no markup, no commercial margin). This is a reciprocal commitment from the Counterparty to Lux, not the other direction.

12. AvaTrade routing: the AvaTrade flow on which Lux holds the direct access (under the framework documented in §VIII) is routed through the joint venue post-SCLA; the Counterparty receives the proportional benefit of the trading volume on the joint-take terms (25–50 bps Lux take, 8 bps internal Counterparty / Alpaca cost stack; see §VIII First Major Prospective Client). Closing of the AvaTrade customer commitment is a separate workstream and not a condition precedent to the SCLA.

13. 8 bps crypto-conversion floor: Lux’s infrastructure enables the joint venue to offer crypto-exchange conversion at 8 basis points (floor); the Counterparty’s existing crypto-conversion pricing is repriced to this floor under the SCLA.

14. Engineering commitment: the Counterparty activates the Lux Warp precompile (canonical address 0x020000000000000000000000000000000005) on every Liquid chain at the next chain-upgrade window. New Liquid chains include the `warpConfig` block in genesis.

15. Litigation hold and document preservation: the Counterparty agrees to preserve all records, communications, and code-review artifacts relating to the licensing posture from 2014 to the SCLA execution date.

16. Independent investigation cooperation: the Counterparty cooperates fully with any independent investigation of officer conduct undertaken by the disinterested directors or their appointed counsel.

17. Whistleblower / non-retaliation protection — Mr. Tridib Nandy (Counterparty VPE; Lux CTO). The Counterparty agrees that Mr. Nandy, who identified and elevated this

matter to the Lux Industries Inc. board through his dual-role luxfi/*-admin visibility and concurrent VPE seat, shall suffer no retaliation, demotion, compensation reduction, or constructive termination from the Counterparty side as a result of his role in this matter. Specifically:

- Mr. Nandy's VPE role and reporting line at the Counterparty shall be preserved on no-less-favourable terms than as of the day preceding the elevation of this matter.
- Indemnification from the Counterparty for any claim asserted against Mr. Nandy personally arising from his role in identifying or escalating the matter, to the maximum extent permitted by applicable corporate law.
- Mutual general release of Mr. Nandy individually from any counterclaim the Counterparty might assert (conditional on full execution of the partnership agreement and ongoing performance).
- Carve-out of Mr. Nandy from any litigation the Counterparty may bring against Lux or any other party arising from this matter.

18. Lux engineering team non-solicit / non- retaliation. The Counterparty agrees, for a period of 36 months post-SCLA execution, not to solicit, recruit, or otherwise induce the departure of any member of the Lux engineering team identified in this matter (a closed list of named individuals to be appended to the SCLA), and not to retaliate against any such individual for any role in the discovery, escalation, or remediation of this matter. General-market hiring through bona fide third-party recruiters or in response to unsolicited applications by the individual is excluded.

19. D&O indemnification of Lux's officers and directors for raising this matter. The Counterparty agrees, by covenant in the SCLA, to release Lux Industries Inc. and its present and former officers and directors (including but not limited to Messrs. Pahlavi, Dupont, Donohue, Nandy, and Kelling) from any direct or derivative claim arising from the Counterparty's use of Lux IP through the SCLA execution date or arising from any party's role in identifying, escalating, or seeking to remediate the matter. The release is mutual and integral to the partnership outcome; refusal of the release by the Counterparty is a deal- breaker.

20. Lux and Hanzo joint warranty of unencumbered title to the licensed IP through the SCLA grant. Lux Industries Inc. warrants that, as of the SCLA execution date, the corporate-identity chain **Monoid LLC (and earlier research vehicles, 2008+) → Hanzo, Inc. (2016) → Lux Partners Limited (IOM) → Lux Industries Inc.** carries unencumbered title to the LEL- and LRL-PR-licensed IP being granted under the SCLA. **Hanzo, Inc.**, as the source-licensor of the Hanzo-origin contribution layer (§IV.A(1B)), additionally grants under the SCLA a parallel royalty-free permanent licence to the Hanzo-source portion of the contributed work product, so that the SCLA regularises both the Lux-licensor and the Hanzo-licensor gaps in a single instrument. Reasonable and customary IP-warranty wording; assignment instruments between successive corporate vehicles to be referenced in the SCLA exhibits.

21. Permanent access for Hanzo, Inc. and Zoo (Zoo Labs Foundation) to the Counterparty's protocol and venue. The SCLA shall grant Hanzo, Inc. and the Zoo Labs Foundation each a **permanent, royalty-free, non-exclusive access right** to the Counterparty's protocol APIs, regulated venue surfaces (ATS / BD / TA), and listed-securities cross-listing mechanisms, for their respective AI / infrastructure (Hanzo) and AI-research / decentralized-science (Zoo) purposes. This access is in addition to Lux's primary equity / cash / reciprocal-licence package and is not a substitute for any other term. The permanent-access grant runs with the LEL grant and survives

any change of control or recapitalisation of the Counterparty.

22. Hanzo, Inc. as named beneficiary of the SCLA. Hanzo, Inc. shall be named as an express third-party beneficiary of the SCLA’s reciprocal-licence, OSS-contribution recognition, and permanent-access provisions, with its own standing to enforce those provisions independent of Lux Industries Inc.’s enforcement rights.

22A. Hanzo, Inc. co-licensor consent and mutual release. Because the Hanzo-source IP contribution layer (§IV.A(1B), Counts I–III, V–VII, XII) is independently held by Hanzo, Inc. and was never licensed to the Counterparty, the SCLA shall include:

- **Hanzo, Inc. as a signing party** to the SCLA in its own name, with parallel grant of a perpetual, royalty-free licence to the Hanzo-source portion of the contributed work product (matching the LEL / LRL–PR perpetual licence ¶2 grant from Lux on the Lux-source portion).
- **Hanzo, Inc. co-licensor consent to the Counterparty’s access** to the Hanzo-side portion of the private IP & patents portfolio described in §VIII (jin, defense, ACI, agent SDK, AGI platform, cloud / blockchain platform, consensus-AI papers, the Hanzo Inc. patent portfolio at §IV.A(4)).
- **Mutual general release of Hanzo, Inc. and its present and former officers, directors, employees, advisors, and consultants** (including but not limited to Mr. Kelling in his Hanzo, Inc. consulting capacity, see §IV.A(1C)) from any direct or derivative claim arising from the Counterparty’s use of Hanzo-source IP through the SCLA execution date.
- **Lux + Hanzo joint release of the Counterparty** on final execution and ongoing performance under the SCLA, in respect of all Counts pleaded jointly or severally in this memorandum. Refusal of the joint release by the Counterparty is a deal-breaker.

The Hanzo, Inc. release on its own behalf, and the parallel Hanzo, Inc. consent for its underlying IP to flow to the Counterparty under the SCLA, are conditions precedent to closing. Without them, the Hanzo-source layer remains independently actionable against the Counterparty (§IV.A(1B); Counts I–III, V–VII, XII), and the partnership outcome cannot be a clean final disposition.

XII. Special Note Re: Mr. Zach Kelling

Mr. Kelling occupies a position requiring particular care in Lux’s partnership and (alternative-path) litigation strategy. The factual context established in §IV(E) and §IV(F) bears directly here:

- Mr. Kelling **departed Lux Industries Inc.** and is **currently Chief Technology Officer of the Counterparty (Liquidity.io) only** (not concurrently a Lux officer or director).
- He was the *remediator* of the pre-Kelling Counterparty technical and security failures (the broken time-price priority matching; the \$40,000 ACH-exploit theft) and the architect of the rebuilt platform that supports the current \$75M / \$1.1B-valuation raise, including personally introducing Alpaca as the clearing BD and developing the Alpaca-integration exchange tech over Q4 2025 (§IV.A(3)).
- **AI-assisted-development mitigating context:** Mr. Kelling and his team conducted the rebuild substantially using AI-assisted development tooling at the rapid pace characteristic of late-2025 / early-2026 AI-augmented engineering. Whether the under-recognition of the Lux

LICENSE markers carried into the imported modules was intentional or (as is more likely given the AI-driven pace) unintentional is a question for fact development; in either case, Mr. Kelling has indicated to Lux that he now sides with the Lux team on the clean precedent that the Lux IP required a commercial licence and that the appropriate disposition is partnership regularisation, not adversarial litigation.

- Mr. Kelling is not the author of the disclosure conduct underlying Counts IX and XI; the duty to disclose the licensing mismatch to investors sat with the Counterparty's CEO and strategy/CSO chain (§IV.D, §V Count V).

Litigation-strategy implications:

- Mr. Kelling should *not* be named as a defendant in any Lux suit. The conduct alleged is not his.
- Mr. Kelling is likely a material witness for Lux on (a) the authorship and ownership of the Lux IP, (b) the absence of any executed commercial license, and (c) the operational-displacement narrative explaining why the mismatch was not raised internally.
- Counsel should consider a litigation-cooperation agreement, witness-protection arrangement, or formal indemnification of Mr. Kelling (as a non-defendant witness) to ensure his continued availability without undue personal exposure.
- In any public communication, Lux should take care not to attribute the misconduct to Mr. Kelling. The narrative should be specific to the Counterparty officers actually responsible.
- In the SCLA itself, Lux should provide for Mr. Kelling's protective treatment, including (a) a release of any claims Lux might assert against him personally, (b) acknowledgment of his historical authorship, and (c) a ROFR / similar mechanism if his Counterparty role generates IP that should flow back to Lux upstream.

Mr. Kelling can serve as a productive intermediary in the Lux–Counterparty negotiation, given his founder lineage with Lux, his current role as Counterparty CTO, his stated alignment with the Lux position on the need for licensing regularisation, and his status as the indispensable key-person on the Counterparty's operational side (§XI ¶10). Mr. Dupont should consider engaging Mr. Kelling directly in informational meetings during the negotiation phase, subject to outside counsel's oversight, with the explicit framing that key-person retention (§XI ¶10) is on the table as part of the partnership outcome.

XIII. Procedural Recommendations

A. Recommended Approach to Counterparty (Partnership-First)

The operative posture is **partnership-first / litigation-as- alternative**. The sequencing the deal team should run:

1. **Quiet inbound to Counterparty CEO (Mr. Choi)** — chair-to-chair / counsel-to-counsel introduction of this matter under a friendly framing, leading with the partnership offer (§X Recommended tier as opener: 25% equity + \$275M cash to \$550M nominal). Frame the conversation as "we identified a licensing mismatch that needs regularisation; here is how we propose to regularise it." Do not lead with Counts or damage ranges.

2. **Parallel quiet investigation track** — continue outside-counsel diligence on the on-disk evidence (commit history, LICENSE-marker travel into the Counterparty tree, current-raise disclosure materials, Sonsurkar / Trombley / Choi communication record) so the leverage is preserved if the partnership track stalls. Litigation hold notice (§XI ¶15) issued early.
3. **Triangulate to Recommended (R, 25%)** — this is the landing zone Lux should anchor on. Floor (10%) is acceptable only with substantial non-equity package (BD/ATS/TA reciprocal license, BaaS access, multi-year cash schedule) that NPV-equivalents the equity gap. Stretch (35%) and Maximum (50%) are upside if the Counterparty wants to preserve cash via additional equity.
4. **Multi-year cash schedule** — the cash component at the lower-equity tiers is large in nominal terms (\$275M at R, \$440M at F) and not deliverable in a single payment from current Counterparty liquidity. Sequence as 30% at close, balance over 24–36 months from operating cash flow, secured by a security interest in Lux’s pledged equity stake or a default-to-additional- equity conversion clause (cap-table dilution mechanically backfills if cash schedule slips).
5. **Mutual releases / symmetric protections** — partnership outcome ships with: §XI ¶0 Kelling key-person retention with mutual releases; §XI ¶17 Nandy whistleblower / non-retaliation; §XI ¶18 Lux engineering team non-solicit; §XI ¶19 D&O indemnification; §XI ¶20 Lux IP warranty. Refusal of any of these is a deal-breaker.
6. **Reciprocal commercial commitments** — the §X tier governance / reciprocal package (board seats, LEL grant, BaaS, joint GTM, co-branding, ATS/BD/TA reciprocal license) is substantively load-bearing, not aesthetic. Each item is what gives the deal forward operating value to Lux beyond the equity stake itself.
7. **Escalation ladder if partnership stalls:** (i) reduce the equity ask one tier (do NOT raise it) to surface whether substance-of-objection is equity-size or something else; (ii) if still stalled, ramp the quiet-investigation track to a sealed filing (preserves leverage, doesn’t go public yet); (iii) if no movement within 90 days from first quiet inbound, file publicly on the alternative-posture Counts.
8. **Communications discipline** — no public statements during partnership-first period. Mr. Pahlavi or Mr. Donohue (GC) clears every external communication. Internal-comms reminder issued to Lux engineering team about confidentiality of pending matter.

Time-to-deal target: 60–90 days from first quiet inbound to executed partnership term sheet, conditional on Counterparty engagement. The current Counterparty raise close is a natural forcing function and should be used as the deal-window anchor.

B. Decision Chain

Decision chain:

1. **Mr. Pahlavi (President & CEO)** — final approval on any executed SCLA, litigation filing, or strategic public communication.
2. **Mr. Dupont (Managing Partner, Strategic Transactions; largest investor)** — deal lead. Conducts negotiation; selects settlement tier; presents executed term sheet to Mr. Pahlavi.

3. **Mr. Donohue (General Counsel)** — legal lead. Drafts SCLA; supervises outside counsel; manages litigation hold; leads pre-suit investigation; prepares SCLA for execution.
4. **Mr. Nandy (CTO; memorandum author)** — operational and technical support. Provides factual development on IP portfolio, licensing record, and Counterparty-side operational context.

Recommended outside engagement:

- **Outside litigation counsel:** a firm with DTSA + patent + Delaware-fiduciary-duty experience (Wilson Sonsini, Cooley, Fenwick & West, or Quinn Emanuel).
- **Valuation firm:** independent fairness opinion on equity-for-IP allocation (Houlihan Lokey, Duff & Phelps / Kroll, or specialist boutique).
- **Investigative firm:** separate firm to investigate Counterparty officer conduct (Kroll Investigations, FTI Consulting, or specialist tech-IP investigators).

Target close: 75 days from Mr. Pahlavi’s approval to authorize the deal team, per the milestone schedule:

Day	Action
0	Mr. Pahlavi approves; outside counsel engaged.
7	Litigation hold notice transmitted to Counterparty + named officers.
14	Investigation firm begins document review.
21	Term sheet delivered to Counterparty at selected tier.
30	Counterparty disinterested-director response received.
45	Investigation report finalized; settlement terms tightened/expanded.
60	Definitive SCLA executed.
75	Closing: cash payment, share issuance, license activation.

XIV. Prayer for Relief (Template)

In any litigation, the complaint’s prayer will request:

1. Compensatory damages on each Count, anticipated \$80M to \$590M on IP claims; \$180M to \$890M with opportunity-cost component.
2. Disgorgement of unjust enrichment from the current and any future raises (rebased to enterprise value, not round size).
3. Reasonable royalty (35 U.S.C. §284; 18 U.S.C. §1836(b)(3)(B)).
4. Exemplary damages 2× on DTSA (18 U.S.C. §1836(b)(3)(C)).
5. Treble damages on patent (35 U.S.C. §284).
6. Statutory damages on copyright (17 U.S.C. §504(c)) at elected maximum.
7. Attorneys’ fees (18 U.S.C. §1836(b)(3)(D); 17 U.S.C. §505; 35 U.S.C. §285; 15 U.S.C. §1117(a)).

8. Preliminary and permanent injunctive relief enjoining further use absent executed commercial license.
9. Pre- and post-judgment interest.
10. Such other and further relief as the Court deems just.

Demand for jury trial on all triable issues.

SCHEDULES

Schedule A — Lux IP Inventory in Production

The full inventory is maintained at the companion license audit. Headline:

- BSD-3 / Apache-2.0 / MIT modules: approximately 25 (no Lux action required).
- LEL modules: `aml`, `broker`, `captable`, `compliance`, `corona`, `crypto`, `fhe`, `hsm`, `maker`, `mpc`, `precompile`, `ringtail`, `threshold`, `transfer`, `treasury`, `warp`.
- LRL-PR modules: `vm`, `staking`, `sampler`, `dex` (Go bindings).
- LGPL modules: `evm`.
- Modules without LICENSE: `cex` and small number of related; counsel should treat as no-rights-granted pending upstream remediation.

Schedule B — Lux IP Pipeline (Available Under SCLA)

- AI inference and compute precompiles (LP-9080 family).
- Advanced ZK rollup substrates.
- Post-quantum hybrid signature schemes (X-Wing + ML-KEM; hybrid ML-DSA + ECDSA).
- Threshold MPC dynamic re-sharing protocols.
- Confidential-execution attestation (TEE / SEV-SNP / TDX binding to consensus).
- White-label partner portal infrastructure.
- Closed `~/work/luxcpp/*` (separate Private-Moat Addendum required).

Schedule C — Banking License Cost-to-Replicate

License	Estimated Cost
US national or state banking charter	\$20–50M
Canadian Schedule II / partnership	\$10–20M
MAS Capital Markets Services + banking (Singapore)	\$5–10M
RBNZ deposit-taker (New Zealand)	\$5–15M
Sweden Finansinspektionen (EU passport)	\$5–10M
UK FCA EMI / Payment Institution	\$2–5M
Switzerland FINMA digital-asset banking	\$5–15M
Hong Kong SFC Type 1 / Type 7 + VATP	\$5–15M
Total direct regulatory + capital	\$57–140M
Plus 5–7 years of timeline + execution risk	

Schedule D — Hanzo Inc. Portfolio Comparables

- Damon Motorcycles — electric / smart-motorcycle company.
- Triller — short-form video / social platform.
- Bellabeat — women’s health wearables.
- Unicorn Gold — precious-metals tokenization platform.
- Lux Network — subject of this memorandum.
- Zoo Labs Foundation / Zoo Network — decentralized AI research network.

Retained-parent-equity range: 10%–40%. The 10% ask sits at the floor of that range.

Schedule E — Mandatory Terms (Lux’s Floor)

See §XI above. Sixteen mandatory terms, summarized:

1. 1 Lux director + 1 independent director seat (closing).
2. 99-year exclusive USA-market license.
3. Weighted-average broad-based anti-dilution + full ratchet on down-rounds.
4. Pro-rata participation rights.
5. Tag-along and co-sale rights.
6. Information rights.
7. Patent grant (current and future Lux patents).
8. Maintenance and support of upstream OSS.
9. Reciprocal permanent license to Counterparty ATS / BD / TA / listed securities / cross-listings.
10. Co-branding for Lux-network chains.

11. OnyxPlus ID verification at Lux internal cost.
12. AvaTrade routing into joint venue.
13. 8 bps crypto-conversion floor.
14. Warp precompile activation at next chain-upgrade.
15. Litigation hold and document preservation.
16. Independent investigation cooperation.

End of memorandum. This document is confidential, privileged, and prepared in anticipation of litigation as well as for board strategy purposes. The proposed settlement framework is non-binding until executed as a definitive Settlement and Cross-License Agreement. Distribution outside Lux Industries Inc., its Board of Directors, the named decision-chain officers (Messrs. Pahlavi, Dupont, Donohue, and Nandy), and engaged outside counsel requires the express written authorization of Mr. Pahlavi and Mr. Donohue.